How far automorphisms of a p-adic absolute Galois group move the ring of integers: to the edge of the log-shell, with an application to Step (v) of IUT IV

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Version 1.1.0, 9 November 2026
doi:10.5281/zenodo.24057859

Abstract

Let p be an odd prime and $K/\mathbb{Q}_p$ a finite extension with ramification index $e \le p-2$. A continuous automorphism α of the absolute Galois group G_K induces, through the mono-anabelian reconstruction, a $\mathbb{Q}_p$-linear automorphism α_+ of K. The map α_+ preserves $\Lambda = \log U_K^1 = \mathfrak{m}_K$ and the log-shell $\mathfrak{m}_K^{1-e}$, but in general not the ring of integers $\mathcal{O}_K$ (Hoshi). We show that for every integer m the $\mathbb{Z}_p$-module spanned by all images of $\mathfrak{m}_K^m$ is $\mathfrak{m}_K^{\mathrm{lo}(m)}$, where $\mathrm{lo}(m)$ is the largest integer $\le m$ congruent to 1 modulo e. So the largest drop of the valuation is $(m-1) \bmod e$: the trivial upper bound coming from the invariance of Λ is attained. In particular the images of $\mathcal{O}_K$ span the whole log-shell, and if $e \ge 2$ some integral element is carried to its boundary.

When $[K : \mathbb{Q}_p]$ is even, or odd with residue degree at least 2, this follows from Kondo's description of the kernel of the trace and Kondo's transvections, from the preservation of traces shown by Hoshi–Nishio, and from automorphisms of free groups that mix the pairs of the Jannsen–Wingberg presentation. In the remaining case, odd degree with total ramification, everything is decided by whether one line in $\Lambda/p\Lambda$, the image of $p\mathbb{Z}_p$, can be moved. To move it we construct, on the quotient of G_K by the third term of the descending p-central series of the wild inertia group, an automorphism that shifts the generator x_1 of the Jannsen–Wingberg presentation in the direction of x_2, and we lift it to G_K by Wingberg's lifting theorem (NSW (7.5.15)). As a by-product, for every finite extension $k/\mathbb{Q}_p$ of odd degree $d \ge 3$ with $p \nmid e$ (more generally, whose maximal tamely ramified extension does not contain the p^2-th roots of unity) there is an automorphism with $\alpha_+(y_1) \ne y_1$, where y_1 corresponds to the first generator. In this range this answers the point left open in Remark 1.10 of Kondo's paper, whether such an element exists in odd degree. It also gives automorphisms that move $(\mathbb{Q}_p)_+ \subset k$.

As an application to the holomorphic hull in Step (v) of the proof of IUT IV, Theorem 1.10, we show that at small places the actual indeterminacies (Ind1), (Ind2) always reach the exact value found in the author's earlier note; its hypotheses $\lambda \in \mathbb{Z}$ and (U) are not needed. We also determine the modules spanned by the images for two types that are not small, the cyclotomic boundary ($e = p-1$, $\mu_p \subset K$) and fields unramified over $\mathbb{Q}_2$: in the first the images of $\mathcal{O}_K$ span the whole log-shell, in the second every automorphism preserves $\mathcal{O}_K$. Among the bad places of the explicit initial Θ-datum D109, the one above 534775081 is of the odd-degree, totally ramified kind. We list remaining questions, among them the determination of the group of induced maps itself. Many of the proofs are checked in Lean 4, with the cited results taken as hypotheses. No claim is made about Corollary 3.12 of IUT III or the *abc* conjecture.

MSC 2020: 11S20 (primary); 11S15, 11S31, 14H25. **Keywords:** mixed-characteristic local fields, outer automorphisms of absolute Galois groups, mono-anabelian geometry, Jannsen–Wingberg presentation, Demuškin groups, log-shell, inter-universal Teichmüller theory.

1 What this note is

1.1 Labels

S a statement about a cited text; P proved here; N checked by an exact computation (scripts in the deposit); H not proved. Many of the P statements are checked in Lean 4, with the cited results and some classical facts taken as hypotheses; Table 2 in Section 12 shows which statements are checked under which hypotheses.

1.2 The question

S Let p be an odd prime and $K/\mathbb{Q}_p$ a finite extension. Continuous automorphisms of G_K need not arise from field automorphisms: $G_{\mathbb{Q}_p}$ has nontrivial outer automorphisms [NSW, pp. 421–423], and for many K the image of $\mathrm{Aut}(K)$ in $\mathrm{Out}(G_K)$ is not a normal subgroup [HN, Kondo]. Such an α still induces, through the mono-anabelian reconstruction, a $\mathbb{Q}_p$-linear automorphism α_+ of K [Hoshi], [Kondo, §0], and α_+ preserves the log-shell. On the other hand, the additive group of the ring of integers $\mathcal{O}_K$ has no group-theoretic reconstruction that works uniformly for all local fields [Hoshi, Rem. 4.3.1(iii)], [HoshiTop, Rem. 2.11.2]. The question of this note is how far $\mathcal{O}_K$ is moved, measured exactly: for each integer m we determine the $\mathbb{Z}_p$-module $S_K(m)$ spanned by all images of $\mathfrak{m}_K^m$ and the largest drop $a_K(m)$ of the valuation (Definition 2.4).

1.3 Background

S For K of even degree, the automorphism φ that Kondo builds from the Jannsen–Wingberg presentation ($\varphi(x_2) = x_2x_1$) moves $(\mathbb{Q}_p)_+ \subset K$ [Kondo, Lemma 1.5]; this is the key step in extending results of Hoshi–Nishio [HN] to non-abelian Galois extensions. For odd degree, Kondo writes in the introduction that it is not clear whether the analogue of Theorem C of that paper holds, and in Remark 1.10:

> "In the case where d_k is odd, a proof analogous to that of Theorem 1.9 would require an element $\alpha \in \mathrm{Aut}(G_k)$ such that $\alpha_+(y_1) \neq y_1$. However, at the time of writing, it is not clear to the author whether or not such an element exists. For this reason, the assumption that d_k is even is imposed in Theorem 1.9." [Kondo, Rem. 1.10]

1.4 What is claimed

(1) P Main theorem (Theorem 6.7). If K is small ($e \leq p-2$), then for every integer m

$$S_K(m) = \mathfrak{m}_K^{\mathrm{lo}(m)}, \qquad a_K(m) = (m-1) \bmod e.$$

In particular the images of $\mathcal{O}_K$ span the log-shell $\mathfrak{m}_K^{1-e}$, and if $e \geq 2$ the image of some integral element reaches its boundary.

(2) P An automorphism moving y_1 in odd degree (Theorem 6.5, Corollary 6.8). If $k/\mathbb{Q}_p$ has odd degree $d \geq 3$ and the integer $s(k)$ of §6.2 is 1 (p odd; this holds whenever $p \nmid e$), there is $\varphi \in \mathrm{Aut}(G_k)$ with $\varphi_+(y_1) \equiv y_1 + y_2$ and $\varphi_+(y_i) \equiv y_i$ ($i \geq 2$) modulo $p\Lambda$. In particular, in this range the element asked for in Kondo's Remark 1.10 exists, and so do automorphisms that move $(\mathbb{Q}_p)_+$.

(3) P The steps leading there: the case of even degree and of odd degree with $f \geq 2$ (Theorem 5.1); in odd degree with total ramification, the reduction of the remaining case to a question about one line ℓ_0, and further about the line of the generator x_1 of the presentation (Theorem 6.1, Proposition 6.2).

(4) P Application to Step (v) of IUT IV (Theorem 7.1, Corollary 7.2). At small places the actual (Ind1), (Ind2) always reach the exact value $p^{\lceil\lambda\rceil-\mathfrak{d}_I}\cdot(R_I)^{\sim}$ of the author's earlier note [Tiers].

(5) P Two types that are not small (Section 8). On the cyclotomic boundary (p odd, $e=p-1$, $\mu_p\subset K$) we determine $S_K(m)$ and $a_K(m)$ for every m (Theorem 8.9). For K unramified over $\mathbb{Q}_2$, every induced map preserves every $\mathfrak{m}_K^m$, and hence $\mathcal{O}_K$ (Theorem 8.11). The tool is that the group of principal units that have p-th roots in the unramified extension of degree p is determined by the group (Lemma 8.1). For small K of odd degree and on the cyclotomic boundary we also determine the union of the images itself (Proposition 8.13).

1.5 What is not claimed

- The group Γ_K of induced maps itself is not determined; only the modules spanned by the images of the lattices $\mathfrak{m}_K^m$ (and, in some cases, the unions of the images) are. Section 10 lists this as a remaining question.
- (1) and (4) concern only small K (p odd, $e\le p-2$). (5) and Remark 5.5 also treat $e=p-1$ and fields unramified over $\mathbb{Q}_2$, but no statement about the lattices is made when $p=2$ and K is ramified, or when p is odd and $e\ge p$.
- The φ of (2) is explicit only modulo the second term of the descending p-central series of the wild inertia group; its existence as an automorphism of G_k rests on Wingberg's lifting theorem (Remark 6.10). (2) is proved only for $s(k)=1$; the case $s(k)\ge 2$, which occurs only when $p\mid e$, is not proved (Remark 6.11). Version 1.0.0 stated it including this case (see the section on changes at the end).
- The upper bound $S_K(m)\subseteq\mathfrak{m}_K^{\mathrm{lo}(m)}$ follows at once from the invariance of Λ and is not new (Proposition 2.5). The automorphisms of free groups that move the pairs (Section 4) are elementary and are the classical constructions for mapping class groups [Kondo, Rem. 2.16, Thm. 2.17], [Ku, §7]; no novelty is claimed for them.
- Nothing about the truth of Corollary 3.12 of IUT III or of the *abc* conjecture. Section 7 is about one local quantity, the holomorphic hull in Step (v) of IUT IV.

2 Setting

2.1 Small local fields

Let p be an odd prime and $K/\mathbb{Q}_p$ a finite extension; write $d=[K:\mathbb{Q}_p]$, e and f for its ramification index and residue degree, π for a uniformizer, v_K for the valuation with $v_K(\pi)=1$, $\mathfrak{m}=\mathfrak{m}_K$, $\mathcal{O}=\mathcal{O}_K$, $U^1=U_K^1=1+\mathfrak{m}$, $q=p^f$, and K_0 for the maximal unramified subfield.

Definition 2.1. K is *small* if $e\le p-2$ (as in [Tiers, Def. 2.1]; this is the case of [IUT4, Prop. 1.2(iii)]).

Lemma 2.2. P *If K is small, then:*

(a) $\mu_p\not\subset K$*; in particular* U^1 *is torsion-free;*

(b) $\log\colon U^1\to\mathfrak{m}$ *is an isomorphism; we put* $\Lambda:=\log U^1=\mathfrak{m}$*;*

(c) the log-shell is $\mathcal{I}_K:=(2p)^{-1}\log\mathcal{O}^\times=p^{-1}\mathfrak{m}=\mathfrak{m}^{1-e}$*;*

(d) $K/\mathbb{Q}_p$ *is tamely ramified and* $\mathrm{Tr}(\mathfrak{m})=p\mathbb{Z}_p$*. More precisely, if* $c\in\mathcal{O}_{K_0}$ *has* $\mathrm{Tr}_{K_0/\mathbb{Q}_p}(c)\in\mathbb{Z}_p^\times$*, then* $pc\in\mathfrak{m}^e$ *and* $\mathrm{Tr}_{K/\mathbb{Q}_p}(pc)\in p\mathbb{Z}_p^\times$*.*

Proof. (a) If $\mu_p \subset K$, then $\mathbb{Q}_p(\mu_p) \subseteq K$; since $\mathbb{Q}_p(\mu_p)/\mathbb{Q}_p$ is totally ramified of degree $p-1$, $p-1 \mid e$, contradicting $e \le p-2$. The torsion of U^1 consists of p-power roots of unity, so it is trivial.

(b) Let $x \in \mathfrak{m}$ and $n \ge 2$. Then $v_K(x^n/n) - v_K(x) \ge (n-1) - e\,v_p(n) > 0$ (if $v_p(n) = k \ge 1$, then $ek \le (p-2)k < p^k - 1 \le n-1$). Similarly $v_K(x^n/n!) - v_K(x) \ge (n-1)(1-e/(p-1)) > 0$. So log and exp are mutually inverse, valuation-preserving maps between U^1 and $\mathfrak{m}$.

(c) $\mathcal{O}^\times = \mu_{q-1} \times U^1$ with $\log \mu_{q-1} = 0$, 2 is a unit, and $p\mathcal{O} = \mathfrak{m}^e$. S See also [IUT4, Prop. 1.2(i)].

(d) As $e < p$, the extension is tame. All conjugates of $x \in \mathfrak{m}$ have absolute value less than 1, so $\mathrm{Tr}(x) \in p\mathbb{Z}_p$. The trace of the residue field extension is surjective, so c exists, and c is a unit. Then $\mathrm{Tr}_{K/\mathbb{Q}_p}(pc) = p\,e\,\mathrm{Tr}_{K_0/\mathbb{Q}_p}(c)$, and e is prime to p. □

2.2 The group Γ_K

S A continuous automorphism α of G_K induces, through the local reciprocity map $\mathrm{rec}_K \colon \widehat{K^\times} \xrightarrow{\sim} G_K^{\mathrm{ab}}$, an automorphism of $\widehat{K^\times}$ that preserves $\mathcal{O}^\times$ and U^1 (the images of the inertia and the wild inertia subgroups). Transporting the automorphism of U^1 by log and extending scalars to $\mathbb{Q}_p$ gives the $\mathbb{Q}_p$-linear map $\alpha_+ \colon K \to K$ induced through the reconstruction [Hoshi], [Kondo, §0]. Inner automorphisms induce the identity. Put $\Gamma_K := \{\alpha_+ \mid \alpha \in \mathrm{Aut}(G_K)\}$, a subgroup of $\mathrm{GL}_{\mathbb{Q}_p}(K)$.

Proposition 2.3. S *Every $\gamma \in \Gamma_K$ satisfies:*

(Γ1) $\gamma(\Lambda) = \Lambda$*; hence $\gamma(p^k\Lambda) = p^k\Lambda$ for $k \in \mathbb{Z}$, and the log-shell is preserved (by definition);*

(Γ2) $\mathrm{Tr}_{K/\mathbb{Q}_p}(\gamma(x)) = \mathrm{Tr}_{K/\mathbb{Q}_p}(x)$ *for $x \in K$ [HN, Lemma 2.3(ii)]. In particular γ preserves the kernel of the trace [Kondo, Lemma 1.2], and hence $W := \Lambda \cap \ker \mathrm{Tr}_{K/\mathbb{Q}_p}$.*

2.3 The size of the movement

Definition 2.4. For an integer m put

$$S_K(m) := \mathrm{span}_{\mathbb{Z}_p}\{\gamma(x) \mid \gamma \in \Gamma_K,\ x \in \mathfrak{m}^m\},$$
$$a_K(m) := m - \min\{v_K(\gamma(x)) \mid \gamma \in \Gamma_K,\ x \in \mathfrak{m}^m \setminus \{0\}\}$$

($S_K(m)$ is the set of finite linear combinations; no closure is taken), and $\mathrm{lo}(m) = \mathrm{lo}_e(m) := e\lfloor (m-1)/e \rfloor + 1$, the largest integer $\le m$ congruent to 1 modulo e. For example $\mathrm{lo}(m) = 1$ for $1 \le m \le e$, and $\mathrm{lo}(0) = 1 - e$.

Proposition 2.5. P *If K is small, then for every m*

$$S_K(m) \subseteq \mathfrak{m}^{\mathrm{lo}(m)}, \qquad 0 \le a_K(m) \le (m-1) \bmod e,$$
$$S_K(m+e) = p\,S_K(m), \qquad a_K(m+e) = a_K(m).$$

Proof. With $q = \lfloor (m-1)/e \rfloor$ we have $\mathfrak{m}^{\mathrm{lo}(m)} = \mathfrak{m}^{qe+1} = p^q\mathfrak{m} = p^q\Lambda$, which is preserved by Γ_K by (Γ1) and contains $\mathfrak{m}^m$. So all images lie in $\mathfrak{m}^{qe+1}$, and $a_K(m) \le m - qe - 1 = (m-1) \bmod e$. The minimum exists because the set of integers in question is bounded below, and $a_K(m) \ge 0$ because the identity lies in Γ_K. The last two statements follow from $\mathfrak{m}^{m+e} = p\,\mathfrak{m}^m$ and $\mathbb{Q}_p$-linearity. □

Remark 2.6. S Hoshi constructs group-theoretically, from the log-shell, a "valuation with an indeterminacy" $\mathrm{ord}_k^{[\mathcal{I}]}(a) = -e \cdot \min\{\nu \in \mathbb{Z} \mid p^\nu a \in \mathcal{I}_k\} + e - 1$ and shows $\mathrm{ord}_k(a) \le \mathrm{ord}_k^{[\mathcal{I}]}(a) < \mathrm{ord}_k(a) + e(\nu_k + 1)$ [HoshiTop, Def. 1.9, Prop. 1.10(ii)] (for small K with $e \ge 2$ one has $\nu_k = 1$). This bounds the drop of the valuation under automorphisms. The bound of Proposition 2.5 is of the same kind. What is new here is that it is attained.

3 The Jannsen–Wingberg presentation and an integral basis

Commutators are $[u,v]=uvu^{-1}v^{-1}$, as in [JW, p. 74].

Theorem 3.1 (Jannsen–Wingberg). S *Let p be odd, $K/\mathbb{Q}_p$ of degree d, and $q=p^f$. Let $\mathcal{G}$ be the profinite group generated by σ,τ with the relation $\sigma\tau\sigma^{-1}=\tau^q$, and let $F(d+1,\mathcal{G})$ be the quotient of the free product of the free profinite group F_{d+1} on $z_0,\dots,z_d$ and $\mathcal{G}$ by the kernel of the maximal pro-p quotient of the closed normal subgroup generated by $z_0,\dots,z_d$; write x_i for the image of z_i. Then there are a positive integer s and an element $r\in F(d+1,\mathcal{G})$, determined by invariants of K, such that $G_K\cong X:=F(d+1,\mathcal{G})/(r)$, with*

$$r=\begin{cases} x_0^{-\sigma}\,(x_0,\tau)^{\beta(\sigma)^{-1}}\,x_1^{p^s}\,[x_1,x_2][x_3,x_4]\cdots[x_{d-1},x_d] & (d\ \textit{even}),\\ x_0^{-\sigma}\,(x_0,\tau)^{\beta(\sigma)^{-1}}\,x_1^{p^s}\,[x_1,x_1'][x_2,x_3]\cdots[x_{d-1},x_d] & (d\ \textit{odd}).\end{cases}$$

Here β is a lifting to $\mathbb{Z}_p^\times$ of the character α by which $\mathcal{G}$ acts on μ_{p^s}, or of its inverse (the conventions differ in the literature), (x_0,τ) is a twisted product formed from x_0 and τ, and x_1' is a word formed from x_1, $\sigma_2=\sigma^{\pi_2}$ and $\tau_2=\tau^{\pi_2}$, where $\pi_\ell\in\widehat{\mathbb{Z}}$ is the idempotent with $\pi_\ell\widehat{\mathbb{Z}}=\mathbb{Z}_\ell$ [JW, pp. 71–75]. The closed normal subgroup generated by $x_0,\dots,x_d$ corresponds to the wild inertia group P_K.

S [JW] write y_1 for the word x_1'; we write x_1', as Kondo does, and keep y_i for elements of Λ (below). Kondo states the same content as follows: there are $x_0'\in\langle\tau,x_0\rangle$, $x_1'\in\langle\sigma,\tau,x_1\rangle$ and a positive integer t with $\sigma x_0\sigma^{-1}=(x_0')^t x_1^{p^s}[x_1,x_2][x_3,x_4]\cdots[x_{d-1},x_d]$ for even d, and with $[x_1,x_1'][x_2,x_3]\cdots[x_{d-1},x_d]$ for odd d [Kondo, Thm. 1.1], based on [NSW, Thm. 7.5.14].

We fix an isomorphism $\phi\colon X\xrightarrow{\sim}G_K$ and identify the x_i with their images. When d is odd, the condition (+) of [JW, 1.2] (f odd and $\alpha(\tau)^{(p-1)/2}\equiv-1 \bmod p$) holds for G_K (see the proof of [JW, Thm. 2]); for small K this is also seen directly: e is odd, and the order $(p-1)/\gcd(e,p-1)$ of $\alpha(\tau)$ modulo p has the same 2-adic valuation as $p-1$.

The images in $\mathcal{G}$ of the twisted products (x,ρ) and $\{x,\rho\}$ are ρ^{π_p} and $\rho^{2\pi_p}$ [JW, p. 74]. The elements ρ occurring in the formulas of [JW] are τ, whose order is prime to p, and elements of the pro-2 group generated by σ_2,τ_2, so these images are trivial. Hence (x_0,τ) and the word x_1' lie in P_K. For $i=1,\dots,d$ put

$$y_i:=\log(\mathrm{rec}_K^{-1}(x_i))\in\Lambda$$

(the image of $x_i\in P_K$ in G_K^{ab} lies in $\mathrm{rec}_K(U^1)$).

Lemma 3.2 (integral basis). P *If K is small, then $y_1,\dots,y_d$ is a $\mathbb{Z}_p$-basis of Λ.*

Proof. We compute the maximal abelian pro-p quotient $X^{\mathrm{ab},(p)}$. The group $F(d+1,\mathcal{G})^{\mathrm{ab},(p)}$ is the free $\mathbb{Z}_p$-module on $\bar\sigma,\bar x_0,\dots,\bar x_d$ (in $\mathcal{G}^{\mathrm{ab}}$ we have $\tau^{q-1}=1$, and the order of τ is prime to p, so $\bar\tau=0$). The commutators in r vanish, and the twisted product becomes a $\mathbb{Z}_p$-multiple of $\bar x_0$; so for some $h\in\mathbb{Z}_p$

$$X^{\mathrm{ab},(p)}=\Big(\mathbb{Z}_p\bar\sigma\oplus\bigoplus_{i=0}^{d}\mathbb{Z}_p\bar x_i\Big)\Big/\mathbb{Z}_p\,(h\,\bar x_0-p^s\bar x_1).$$

If $h\in p\mathbb{Z}_p$ (including $h=0$), the relation vector is divisible by p^k with $k=\min(v_p(h),s)\ge1$, and the quotient has torsion of order p^k. On the other hand, by local class field theory $G_K^{\mathrm{ab},(p)}\cong(K^\times)_p^\wedge\cong\mathbb{Z}_p\times U^1$, which is torsion-free by Lemma 2.2(a). So $h\in\mathbb{Z}_p^\times$, $\bar x_0=h^{-1}p^s\bar x_1$, and $\bar\sigma,\bar x_1,\dots,\bar x_d$ is a basis of $X^{\mathrm{ab},(p)}$. The image of P_K is spanned by $\bar x_0,\dots,\bar x_d$, that is, it is the direct summand spanned by $\bar x_1,\dots,\bar x_d$, and under local class field theory it corresponds to U^1. Transport by log. □

Remark 3.3. S Kondo notes that $y_1, \dots, y_d$ is a basis over $\mathbb{Q}_p$ [Kondo, §1]. Lemma 3.2 makes it integral, and all arguments below take place on this integral basis.

Lemma 3.4. S P *Let K be small and $d \geq 2$. By Kondo [Kondo, Thm. 1.3],*

$$\ker \mathrm{Tr}_{K/\mathbb{Q}_p} = \begin{cases} \langle y_1, y_3, y_4, \dots, y_d \rangle_{\mathbb{Q}_p} & (d \text{ even}), \\ \langle y_2, y_3, \dots, y_d \rangle_{\mathbb{Q}_p} & (d \text{ odd}). \end{cases}$$

Let $y_* = y_2$ for even d and $y_* = y_1$ for odd d. Then (with Lemma 3.2) $W = \Lambda \cap \ker \mathrm{Tr}$ is the $\mathbb{Z}_p$-span of the y_i other than y_*, $\Lambda = \mathbb{Z}_p y_* \oplus W$, and $\mathrm{Tr}(y_*) \in p\mathbb{Z}_p^\times$.

Proof. If $x = \sum c_i y_i \in \Lambda$ ($c_i \in \mathbb{Z}_p$) has trace 0, then $c_* \mathrm{Tr}(y_*) = 0$. If $\mathrm{Tr}(y_*) = 0$, then $\mathrm{Tr}(\Lambda) = 0$, contradicting Lemma 2.2(d); so $c_* = 0$. Also $\mathrm{Tr}(\Lambda) = \mathbb{Z}_p \mathrm{Tr}(y_*) = p\mathbb{Z}_p$. □

4 Automorphisms moving the pairs of the presentation

4.1 From automorphisms of free groups to automorphisms of G_K

Lemma 4.1. P *Let ψ be an automorphism of $F_{d+1} * \mathcal{G}$ that fixes $\mathcal{G}$ pointwise, fixes z_0, z_1, and sends each z_i ($i \geq 2$) to a word in $z_1, \dots, z_d$, its inverse having the same form. If ψ fixes, as an element, the product of commutators $[z_1, z_2][z_3, z_4] \cdots [z_{d-1}, z_d]$ of the relation (for odd d, $[z_2, z_3] \cdots [z_{d-1}, z_d]$), then ψ induces an automorphism α of $X \cong G_K$, and α_+ acts on $\Lambda = \bigoplus \mathbb{Z}_p y_i$ by the matrix of the abelianization $z_i \mapsto \psi(z_i)^{\mathrm{ab}}$ on $\bar{z}_1, \dots, \bar{z}_d$.*

Proof. ψ preserves the normal subgroup generated by the z_i and the kernel of its maximal pro-p quotient (a characteristic subgroup), so it induces an automorphism of $F(d+1, \mathcal{G})$. It fixes σ, τ, x_0, x_1, hence also the word x_1' and the twisted product, which are formed from them. By assumption r is fixed as an element, so (r) is preserved; the same holds for the inverse, and we obtain an automorphism of X. The second statement follows from the definition of α_+ (§2.2) and the identification of Lemma 3.2. □

4.2 Within one pair, and adjacent pairs

Lemma 4.2. P *In a free group:*

(i) $a \mapsto ab$ (b fixed) and $b \mapsto ba$ (a fixed) are automorphisms that fix $[a, b]$; their inverses multiply by the inverse on the right;

(ii) with $D = [c, d]$, the map $a \mapsto c$, $b \mapsto d$, $c \mapsto D^{-1}aD$, $d \mapsto D^{-1}bD$ is an automorphism, with inverse $a \mapsto CcC^{-1}$, $b \mapsto CdC^{-1}$, $c \mapsto a$, $d \mapsto b$ where $C = [a, b]$; it fixes $[a, b][c, d]$ and swaps the two pairs on the abelianization.

Proof. (i) $[ab, b] = ab\,b\,b^{-1}a^{-1}b^{-1} = [a, b]$ and $[a, ba] = a\,ba\,a^{-1}a^{-1}b^{-1} = [a, b]$. (ii) The product of the commutators of the images is $[c, d] \cdot D^{-1}[a, b]D = [a, b]D$. The inverse is checked by substitution. □

4.3 Even degree: mixing the first pair with the others

In the free group on four free generators a, b, c, d (in this subsection d is the name of a generator, not the local degree) put $R = [a, b]$, $Q = [c, d]$, $\delta_2 = RQ$, $k = b^{-1}cdc^{-1}$, and consider

$$\begin{aligned} Z &\colon a \mapsto ak, \quad b \mapsto k^{-1}bk, \quad c \mapsto k^{-1}c, \quad d \mapsto d; \\ H &\colon a \mapsto RcR^{-1}, \quad b \mapsto RdR^{-1}, \quad c \mapsto a, \quad d \mapsto b; \\ J &\colon a \mapsto aba^{-1}, \quad b \mapsto a^{-1}, \quad c \mapsto c, \quad d \mapsto d. \end{aligned}$$

Lemma 4.3. P *Z, H, J are automorphisms that fix δ_2 as an element. With $\Psi = J \circ H$ and $M = \Psi \circ Z \circ \Psi^{-1}$ we have $M(a) = a$ and $M(\delta_2) = \delta_2$.*

Proof. $Z(k) = k$ by substitution $(Z(b)^{-1}Z(c)\,d\,Z(c)^{-1} = k^{-1}b^{-1}k \cdot k^{-1}c\,d\,c^{-1}k = k^{-1} \cdot k \cdot k)$, and the inverse of Z is $a \mapsto ak^{-1}$, $b \mapsto kbk^{-1}$, $c \mapsto kc$, $d \mapsto d$. With $u = cdc^{-1} = bk$,

$$\begin{aligned}[ak, k^{-1}bk]\,[k^{-1}c, d] &= aba^{-1}k^{-1}b^{-1}k \cdot k^{-1}cdc^{-1}kd^{-1} \\ &= aba^{-1}k^{-1}(b^{-1}u)kd^{-1} = aba^{-1}kd^{-1} = [a, b][c, d].\end{aligned}$$

The inverse of H is $a \mapsto c$, $b \mapsto d$, $c \mapsto Q^{-1}aQ$, $d \mapsto Q^{-1}bQ$, and $H(R) = RQR^{-1}$, $H(Q) = R$, so $H(\delta_2) = \delta_2$. The inverse of J is $a \mapsto b^{-1}$, $b \mapsto bab^{-1}$, and $[aba^{-1}, a^{-1}] = [a, b]$. Finally $\Psi(d) = J(b) = a^{-1}$, so $\Psi^{-1}(a) = d^{-1}$, $Z(d^{-1}) = d^{-1}$, and $M(a) = \Psi(d^{-1}) = a$. □

On the abelianization put the alternating form with $\langle a, b\rangle = \langle c, d\rangle = -1$, $\langle b, a\rangle = \langle d, c\rangle = 1$ and 0 on the other pairs of basis elements, and write $T_v(x) = x + \langle x, v\rangle\, v$. The map $b \mapsto ba$ of Lemma 4.2(i) induces T_a, and $c \mapsto cd^{-1}$ induces T_d (note $[cd^{-1}, d] = [c, d]$). The abelianization of Z is $a \mapsto a + d - b$, $c \mapsto c + b - d$, which is T_{d-b}. The abelianization of Ψ is $a \mapsto c$, $b \mapsto d$, $c \mapsto b$, $d \mapsto -a$, which preserves the form; from $\Psi(d - b) = -a - d$ we get

$$M^{\mathrm{ab}} = T_{a+d}.$$

As $\langle a, d\rangle = 0$,

$$U_d := T_{a+d}\,T_a^{-1}\,T_d^{-1}, \qquad U_d(x) = x + \langle x, a\rangle\, d + \langle x, d\rangle\, a.$$

The automorphism of the free group that induces U_d (the composite of M with the inverses of $b \mapsto ba$ and $c \mapsto cd^{-1}$) fixes a and δ_2.

Proposition 4.4. P *Let K be small, and put on Λ the alternating form for which, in each pair—$(y_1, y_2), (y_3, y_4), \ldots$ for even d, and $(y_2, y_3), (y_4, y_5), \ldots$ for odd d—the first element u and the second v satisfy $\langle u, v\rangle = -1$, and which is 0 on the other pairs of basis elements.*

(a) Let d be even, $a = y_1$, $b = y_2$ and $W' = \langle y_3, \ldots, y_d\rangle_{\mathbb{Z}_p}$. Then $T_a \in \Gamma_K$, and for $w \in \{\pm y_3, \ldots, \pm y_d\}$ the map $U_w(x) = x + \langle x, a\rangle\, w + \langle x, w\rangle\, a$ lies in Γ_K.

(b) Let d be odd. For each pair (u, v) the maps $v \mapsto v + u$ and $u \mapsto u + v$, and the swap of adjacent pairs, lie in Γ_K. All of them fix y_1.

Proof. (a) T_a comes from $x_2 \mapsto x_2x_1$ (Kondo's φ). U_{y_4} and U_{-y_3} are obtained by applying the construction of Lemma 4.3 to the first two pairs $(x_1, x_2), (x_3, x_4)$: applying J to the second pair gives $d \mapsto -c$ on the abelianization, and conjugating gives U_{-c}; indeed, if Ψ' preserves the form and fixes a, then $\Psi' U_w \Psi'^{-1} = U_{\Psi'(w)}$. The other pairs are reached by conjugating with the swaps of Lemma 4.2(ii), which fix x_1, x_2. Since $U_{-w} = U_w^{-1}$, both signs occur. Every automorphism of the free group used here fixes x_0, x_1 and the product of commutators, so the induced maps lie in Γ_K by Lemma 4.1. (b) Apply Lemma 4.2 to $x_2, \ldots, x_d$ and use Lemma 4.1. □

Remark 4.5. S Part (b) is in substance contained in Kondo's result, obtained through mapping class groups, that the induced maps include a subgroup that fixes y_1 and acts on $\langle y_2, \ldots, y_d\rangle$ as $\mathrm{Sp}_{2g}(\mathbb{Z}_p)$ [Kondo, Rem. 2.18, proof of Thm. 2.19] (there $k/\mathbb{Q}_p$ is assumed Galois). The automorphism $x_N \mapsto x_N x_{N-1}$ used by NSW [NSW, p. 421], which Kumpitsch calls a Jannsen–Wingberg automorphism [Ku, §3.2.2], is in odd degree one of the maps in (b). The maps U_w of (a) are presumably also obtainable by Kondo's method [Kondo, Rem. 2.21]; here they are given by explicit words. All of this is elementary, and no novelty is claimed.

5 The exact value

Theorem 5.1. P *Let K be small and m an integer, and suppose one of the following:*

(a) d is even;

(b) d is odd and $f \geq 2$;

(c) d is odd, $f = 1$ (so $d = e$), and $e \nmid m$.

Then

$$S_K(m) = \mathfrak{m}^{\mathrm{lo}(m)}, \qquad a_K(m) = (m-1) \bmod e.$$

In particular in cases (a) and (b) the images of $\mathcal{O}_K$ span the log-shell $\mathcal{I}_K = \mathfrak{m}^{1-e}$, and $v_K(\gamma(x)) = 1 - e$ for some $x \in \mathcal{O}_K$ and $\gamma \in \Gamma_K$. In case (a), moreover, $S_K(m)$ is spanned by at most d images of a single element.

If $e = 1$, then $\mathrm{lo}(m) = m$ and the statement is trivial (invariance of $\Lambda = p\mathcal{O}$); if $d = 1$, then $K = \mathbb{Q}_p$. From now on assume $e \geq 2$.

5.1 Even degree

Lemma 5.2. P *Let d be even. If $x \in \Lambda$ and $\mathrm{Tr}(x)$ generates $p\mathbb{Z}_p$, then d elements of $\Gamma_K x$ form a $\mathbb{Z}_p$-basis of Λ.*

Proof. In the notation of Proposition 4.4(a) write $x = \alpha a + ub + w_0$ with $\alpha, u \in \mathbb{Z}_p$ and $w_0 \in W'$. By Lemma 3.4, $\mathrm{Tr}(x) = u\,\mathrm{Tr}(b)$, so u is a unit. As $\langle x, a\rangle = u\langle b, a\rangle = u$,

$$T_a x - x = ua, \qquad U_w x - x = uw + \langle x, w\rangle\, a \quad (w = y_3, \dots, y_d).$$

Take the d elements x, $T_a x$, $U_{y_3}x, \dots, U_{y_d}x$. Their differences give ua, and after removing the a-component, $uy_3, \dots, uy_d$; subtracting suitable combinations of these from x leaves ub. As u is a unit, the d elements span Λ, so they form a basis. □

Proof of Theorem 5.1(a). Let $q = \lfloor (m-1)/e \rfloor$, let c be as in Lemma 2.2(d), and put $x_m := p^{q+1}c$. Then $v_K(x_m) = (q+1)e \geq m$, so $x_m \in \mathfrak{m}^m$. The trace of pc generates $p\mathbb{Z}_p$, so by Lemma 5.2 some d images of pc span Λ; multiplying by p^q, d images of x_m span $p^q\Lambda = \mathfrak{m}^{\mathrm{lo}(m)}$. With Proposition 2.5, $S_K(m) = \mathfrak{m}^{\mathrm{lo}(m)}$. If these finitely many images all lay in $\mathfrak{m}^{\mathrm{lo}(m)+1}$, so would their span; hence one of them has valuation $\mathrm{lo}(m)$, and $a_K(m) = m - \mathrm{lo}(m) = (m-1) \bmod e$. □

5.2 Odd degree

Lemma 5.3. P *Let d be odd. If $w \in W \setminus pW$, then $\mathrm{span}_{\mathbb{Z}_p}(\Gamma_K w) = W$.*

Proof. As $w \notin pW$, for some pair (u, v) the u-component α or the v-component β of w is a unit. If α is a unit, the image of w under $u \mapsto u + v$ differs from w by αv, so v lies in the span of the images of w; the image of v under $v \mapsto v + u$ differs from v by u. If β is a unit, exchange the roles. The span is stable under Γ_K, so by the swaps of pairs (Proposition 4.4(b)) it contains the basis elements of all pairs. The reverse inclusion is (Γ2). □

Lemma 5.4. P *Let d be odd, $1 \leq r \leq e$ and $f(e+1-r) \geq 2$. Then $S_K(r) = \Lambda$.*

Proof. We have $\mathfrak{m}^r \supseteq p\Lambda = \mathfrak{m}^{e+1}$ and $\dim_{\mathbb{F}_p}(\mathfrak{m}^r/p\Lambda) = f(e+1-r) \geq 2$. The trace maps $p\Lambda$ into $p^2\mathbb{Z}_p$, so it induces $\mathfrak{m}^r/p\Lambda \to p\mathbb{Z}_p/p^2\mathbb{Z}_p$, which is surjective because of $pc \in \mathfrak{m}^e \subseteq \mathfrak{m}^r$ (Lemma 2.2(d)). Its kernel is nonzero, so there is $z \in \mathfrak{m}^r \setminus p\Lambda$ with $\mathrm{Tr}(z) \in p^2\mathbb{Z}_p$. Put $t = \mathrm{Tr}(z)/\mathrm{Tr}(pc) \in p\mathbb{Z}_p$ and $w = z - t\,pc$. Then $w \in \mathfrak{m}^r \cap \ker \mathrm{Tr} \subseteq W$ and $w \equiv z \not\equiv 0 \pmod{p\Lambda}$; in particular $w \notin pW$, so $S_K(r) \supseteq W$ by Lemma 5.3. Also $pc \in S_K(r)$ has trace generating $p\mathbb{Z}_p$, so $S_K(r) \supseteq \Lambda$ by Lemma 3.4. The reverse inclusion is (Γ1). □

Proof of Theorem 5.1(b)(c). Write $m = qe + r$ with $1 \le r \le e$. In (b), $f \ge 2$, so the hypothesis of Lemma 5.4 holds for every r; in (c), $e \nmid m$, that is, $r \le e-1$, so $e+1-r \ge 2$ and the hypothesis holds again. As $\mathfrak{m}^m = p^q \mathfrak{m}^r$, we get $S_K(m) = p^q S_K(r) = p^q \Lambda = \mathfrak{m}^{\mathrm{lo}(m)}$. For $a_K(m)$ argue as in (a): not all images can lie in $\mathfrak{m}^{\mathrm{lo}(m)+1}$. □

Remark 5.5. P If $e = p-1$ (so d is even) and $\mu_p \not\subset K$, Theorem 5.1(a) still holds. Then exp does not converge on all of $\mathfrak{m}$, but log and exp are mutually inverse between U^2 and $\mathfrak{m}^2$, and $\log(1+\pi z) \equiv \pi(z + cz^p) \pmod{\mathfrak{m}^2}$ with $c = \pi^{p-1}/p \in \mathcal{O}^\times$. If the map $z \mapsto z + \bar{c}z^p$ of the residue field had a nonzero kernel, the reduction $1 + \bar{c}Z^{p-1}$ of $\Phi_p(1+\pi Z)/p$ would have a root, and by Hensel's lemma $\mu_p \subset K$, a contradiction. So $\log U^1 = \mathfrak{m}$, and the arguments from Lemma 3.2 on go through.

type of K	m	$S_K(m)$	$a_K(m)$	reference
d even	all	$\mathfrak{m}^{\mathrm{lo}(m)}$	$(m-1) \bmod e$	Thm. 5.1(a)
d odd, $f \ge 2$	all	$\mathfrak{m}^{\mathrm{lo}(m)}$	$(m-1) \bmod e$	Thm. 5.1(b)
$d = e$ odd	$e \nmid m$	$\mathfrak{m}^{\mathrm{lo}(m)}$	$(m-1) \bmod e$	Thm. 5.1(c)
$d = e$ odd	$e \mid m$	$\mathfrak{m}^{m+1-e} = \mathfrak{m}^{\mathrm{lo}(m)}$	$e-1$	Cor. 6.6, Thm. 6.7 (§6)

Table 1: The module spanned by the images of $\mathfrak{m}^m$ (K small, $e \ge 2$; $d = e$ odd means odd degree and total ramification).

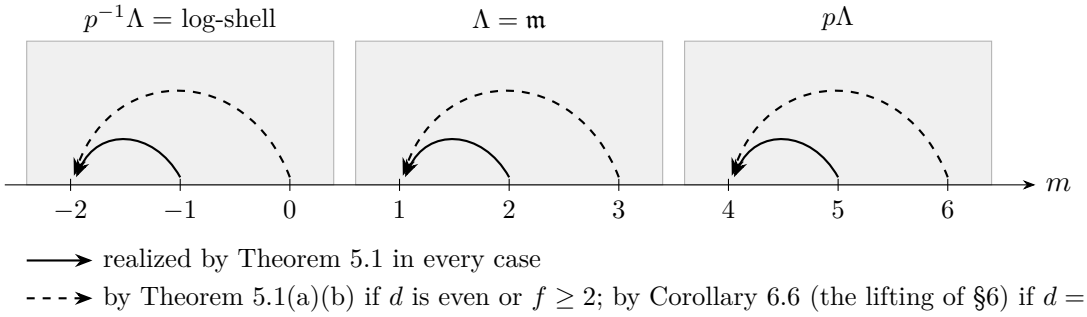


Figure 1: The case $e = 3$. Within each block of three exponents (a layer $p^q\Lambda$), the images of $\mathfrak{m}^m$ span up to the outermost lattice $\mathfrak{m}^{\mathrm{lo}(m)}$ of that block. The moves from exponents divisible by e (including $\mathcal{O}_K = \mathfrak{m}^0$), drawn dashed, need the lifting of Section 6 exactly when $d = e$ is odd.

6 Moving the first generator in odd degree

6.1 The dichotomy and a reduction

In this subsection K is small, $d = e \ge 3$ is odd and $f = 1$. By the periodicity of Proposition 2.5 it suffices to treat $m = 0$. Put

$$\ell_0 := (p\mathcal{O} + p\Lambda)/p\Lambda \subset \Lambda/p\Lambda.$$

As $p\mathcal{O} = \mathfrak{m}^e \supseteq p\mathfrak{m} = p\Lambda$ and $p\mathcal{O}/p\mathfrak{m} \cong \mathcal{O}/\mathfrak{m} = \mathbb{F}_p$, the line ℓ_0 is one-dimensional over $\mathbb{F}_p$, spanned by the image of p (the image of $\log U^1_{\mathbb{Q}_p} = p\mathbb{Z}_p$).

Theorem 6.1. P *Exactly one of the following holds.*

(i) Every $\gamma \in \Gamma_K$ preserves ℓ_0. Then $\gamma(\mathcal{O}_K) = \mathcal{O}_K$ for every γ, and for $e \mid m$ we have $S_K(m) = \mathfrak{m}^m$ and $a_K(m) = 0$.

(ii) Some $\gamma \in \Gamma_K$ does not preserve ℓ_0. Then for $e \mid m$ we have $S_K(m) = \mathfrak{m}^{m+1-e}$ and $a_K(m) = e-1$; in particular the images of $\mathcal{O}_K$ span the log-shell $\mathfrak{m}^{1-e}$.

Proof. The preimage of ℓ_0 under $\Lambda \to \Lambda/p\Lambda$ is $p\mathcal{O}$. In case (i), γ preserves $p\mathcal{O}$, hence $\mathcal{O}$. In case (ii), for some γ we have $\gamma(p) \not\equiv bp \pmod{p\Lambda}$ for all $b \in \mathbb{Z}_p$. By (Γ2), $\mathrm{Tr}(\gamma(p)) = \mathrm{Tr}(p)$, so $w := \gamma(p) - p \in W$, and $w \notin p\Lambda$, that is, $w \notin pW$. By Lemma 5.3, $\operatorname{span}\Gamma_K(w) = W$. The module $S_K(e) \supseteq \operatorname{span}\Gamma_K(p)$ contains w and p, hence Λ by Lemma 3.4, and $S_K(e) = \Lambda$ by (Γ1). Dividing by p gives $S_K(0) = p^{-1}\Lambda = \mathfrak{m}^{1-e}$. □

For each isomorphism $\phi\colon X \xrightarrow{\sim} G_K$ (Theorem 3.1) let $L_\phi \subset \Lambda/p\Lambda$ be the line spanned by the image of $y_1^\phi := \log \mathrm{rec}_K^{-1}(\phi(x_1))$ (by Lemma 3.2, $y_1^\phi \notin p\Lambda$).

Proposition 6.2. P *Case (i) of Theorem 6.1 holds if and only if $L_\phi = \ell_0$ for every ϕ. In particular, if $L_\phi \neq \ell_0$ for one ϕ, that is, if $y_1^\phi \notin \mathbb{Z}_p^\times p + p\Lambda$, then case (ii) holds.*

Proof. For $\alpha \in \mathrm{Aut}(G_K)$, $\alpha\circ\phi$ is again an isomorphism and $L_{\alpha\circ\phi} = \alpha_+(L_\phi)$; conversely any two isomorphisms differ by an element of $\mathrm{Aut}(G_K)$. So $\{L_\phi\}$ is one Γ_K-orbit. Suppose $L_\phi \neq \ell_0$ for some ϕ. As $\Lambda/p\Lambda = \mathbb{F}_p\bar{y}_1 \oplus W/pW$ (Lemma 3.4, in the basis of ϕ), we can write $\ell_0 = \mathbb{F}_p(\bar{y}_1 + \bar{w})$ with $\bar{w} \in W/pW \setminus \{0\}$. The alternating form on W/pW is nondegenerate, so some basis element v of a pair has $\langle \bar{w}, v\rangle \neq 0$. The transvection T_v (or its inverse) of Proposition 4.4(b) fixes $\bar{y}_1$ and sends $\bar{y}_1 + \bar{w}$ to $\bar{y}_1 + \bar{w} + \langle \bar{w}, v\rangle\, v$, which is not an $\mathbb{F}_p$-multiple of $\bar{y}_1 + \bar{w}$ (the coefficient of $\bar{y}_1$ stays 1). So case (ii) holds. Conversely, if $L_\phi = \ell_0$ for every ϕ, the orbit $\{L_\phi\}$ is $\{\ell_0\}$, so case (i) holds. □

6.2 Two tools

From now on $k/\mathbb{Q}_p$ is any finite extension of odd degree $d \geq 3$ (not necessarily small). Let $\mathcal{G} = \mathrm{Gal}(k_{\mathrm{tr}}/k)$ be the tame quotient, $V = V_k$ the wild inertia group, and $V^{(1)} = V$, $V^{(i+1)} = (V^{(i)})^p[V^{(i)}, V]$ its descending p-central series (the numbering of [NSW]; the X^i of [JW] correspond to $V^{(i+1)}$). As d is odd, $\mu_p \not\subset k$, so the character α by which $\mathcal{G}$ acts on μ_{p^s} is nontrivial modulo p.

Here $s = s(k)$ is the integer such that the maximal tamely ramified extension k_{tr} of k contains exactly p^s roots of unity of p-power order (the s of the Jannsen–Wingberg presentation [NSW, p. 419]; $s \geq 1$ since $\mu_p \subset k_{\mathrm{tr}}$). If $p \nmid e(k)$, then $s(k) = 1$. Indeed, if $s \geq 2$, then $k(\mu_{p^2}) \subseteq k_{\mathrm{tr}}$ is tamely ramified over k, so its ramification index over $\mathbb{Q}_p$ is $e(k)$ times a number prime to p; on the other hand it is divisible by the ramification index $p(p-1)$ of $\mathbb{Q}_p(\mu_{p^2})$, so $p \mid e(k)$. In particular $s(K) = 1$ for small K.

Theorem 6.3 (Wingberg's lifting theorem)**.** S *Let $p \neq 2$ and $s(k) = 1$, and let ψ_3 be an automorphism of $G_k/V^{(3)}$ that induces the identity on G_k/V. Then there is an automorphism φ of G_k that coincides with ψ_3 modulo $V^{(2)}$ [Wi, Satz 2], [NSW, Thm. 7.5.15], [Nishio26, Prop. 2.6].*

S NSW use this theorem to construct nontrivial outer automorphisms of $G_{\mathbb{Q}_p}$ [NSW, pp. 421–423]. Nishio uses it to construct finite subgroups of $\mathrm{Aut}(G_k)$ acting on the pairs $(x_2, x_3), \ldots$ [Nishio26, Prop. 2.6]. Here we construct an automorphism that preserves the relation itself at the third level and lift it by this theorem.

S P **A remark on s.** NSW state (7.5.15) with the descending p-central series, after noting that Wingberg's statement was not correct as stated. Nishio states the same theorem, for $s = s(k)$, with the descending p^s-central series $H^{p,i+1,s} = (H^{p,i,s})^{p^s}[H, H^{p,i,s}]$ [Nishio26, Prop. 2.6]. For $s = 1$ the two agree, and this note uses only this case. For $s \geq 2$ the form with the descending p-central series cannot be used. For example (in even degree), for $k = \mathbb{Q}_p(\zeta_{p^2})$, where $s = 2$, the map $x_1 \mapsto x_1x_2$ (the other generators fixed) gives an automorphism of

$G_k/V^{(3)}$ that is the identity on G_k/V (as $x^{p^2} \in V^{(3)}$ and $[x_1x_2, x_2] \equiv [x_1, x_2] \pmod{V^{(3)}}$). An automorphism of G_k that coincides with it modulo $V^{(2)}$ would induce the same map on G_k^{ab}/p (as $V^{(2)} \subseteq G_k^p[G_k, G_k]$), and so would move the line in G_k^{ab}/p spanned by the image of the torsion subgroup; but automorphisms of G_k preserve this line, so no such automorphism exists.

Notation. Write $F = F(d+1, \mathcal{G}) = P \rtimes \mathcal{G}$, where P is the free pro-p operator group with operator domain $\mathcal{G}$ on the free generators $x_0, \dots, x_d$ [JW, p. 74]. Then $G_k = F/\langle\langle r \rangle\rangle$, and V is the image of P. Let $P^{(i)}$ be the descending p-central series of P, and put $M_1 = P/P^{(2)}$, $M_2 = P^{(2)}/P^{(3)}$. The group $\mathcal{G}$ acts from the left by conjugation, $g \cdot x = gxg^{-1}$ (the convention $y^x = xyx^{-1}$ of [JW]), making both into modules over $R := \mathbb{F}_p[[\mathcal{G}]]$; M_1 is free on $\bar{x}_0, \dots, \bar{x}_d$. As p is odd, the p-th power gives an R-linear map $M_1 \to M_2$, and the commutator gives an alternating bilinear map $B\colon M_1 \times M_1 \to M_2$ with $B(gu, gv) = g\,B(u, v)$ for $g \in \mathcal{G}$. Let $\varepsilon\colon R \to \mathbb{F}_p$ be the augmentation.

Proposition 6.4 (the linear part of the relation)**.** S

(i) The relation r and the Jannsen–Wingberg word x_1' lie in P (Section 3).

(ii) There is $\delta \in R$ with $x_1' \equiv \delta\,\bar{x}_1 \pmod{P^{(2)}}$ [JW, 2.9].

(iii) The image of the twisted product (x_0, τ) in M_1 is $e'\bar{x}_0$, where e' is the idempotent of $\mathbb{F}_p[[\langle\tau\rangle]]$ for the component on which τ acts by $\beta(\tau)^{-1}$, and e' commutes with σ [JW, 2.3 and formula (2) on p. 81]. Hence e' is central in R, and the image of r in M_1 is $\theta\,\bar{x}_0$ with $\theta := -\sigma + \beta(\sigma)^{-1}e'$ (modulo p).

6.3 The construction

Theorem 6.5. P *Let p be an odd prime and $k/\mathbb{Q}_p$ a finite extension of odd degree $d \geq 3$ with $s(k) = 1$ (which always holds if $p \nmid e$). Fix Jannsen–Wingberg generators and put $y_i = \log \mathrm{rec}_k^{-1}(x_i)$. Then there is an automorphism φ of G_k with*

$$\varphi_+(y_1) \equiv y_1 + y_2, \qquad \varphi_+(y_i) \equiv y_i \quad (2 \leq i \leq d) \pmod{p\Lambda},$$

where $\Lambda = \log U_k^1$. In particular $\varphi_+(y_1) \neq y_1$.

Proof. We use the notation of §6.2.

(1) The effect of adjusting x_0. For $z \in P^{(2)}$, replacing x_0 by x_0z changes r modulo $P^{(3)}$ into $r \cdot \theta\bar{z}$. Indeed, z is central modulo $P^{(3)}$, so the x_0-part $x_0^{-\sigma}(x_0, \tau)^{\beta(\sigma)^{-1}}$ of r changes linearly in z, with the coefficient θ of Proposition 6.4(iii); the remaining part $x_1^p[x_1, x_1'][x_2, x_3]\cdots$ does not involve x_0.

(2) The character and the ideal. θR is a two-sided ideal of R of codimension 1. Indeed, e' is a central idempotent and $\theta R = (1-e')\sigma R + (b-\sigma)e'R = (1-e')R + (b-\sigma)e'R$ with $b = \beta(\sigma)^{-1}$ mod p; in $e'R = e'\,\mathbb{F}_p[[\langle\sigma\rangle]]$ the element $b - \sigma$ generates the kernel of the character $\sigma \mapsto b$. So θR is the kernel I_0 of the augmentation of the character $\chi_0\colon \sigma \mapsto b$, $\tau \mapsto \beta(\tau)^{-1}$ (modulo p). Since β is a lifting of α or of α^{-1} (the conventions differ: [JW, pp. 74–75], [NSW, p. 420]), $\chi_0 = \alpha^{\pm 1}$ mod p. As d is odd, $\mu_p \not\subset k$, so χ_0 is nontrivial. Extend $g^\dagger := \chi_0(g)g^{-1}$ to an anti-automorphism of R. For $g \in \mathcal{G}$ we have $B(gu, v) = g\,B(u, g^{-1}v) \equiv \chi_0(g)B(u, g^{-1}v) = B(u, g^\dagger v) \pmod{I_0M_2}$, hence

$$B(au, v) \equiv B(u, a^\dagger v) \pmod{I_0M_2} \qquad (a \in R).$$

(3) The map. Choose $\rho_0 \in \mathcal{G}$ with $\chi_0(\rho_0) \neq 1$ and put $c := (\rho_0 - \chi_0(\rho_0))/(1 - \chi_0(\rho_0)) \in R$. Then $\varepsilon(c) = 1$ and $c \in I_0$. Put $\mu := -c^\dagger(\delta - \delta^\dagger)$ and $\nu := -c^\dagger\delta c$. As $\varepsilon(c^\dagger) = \chi_0(c) = 0$, we have $\varepsilon(\mu) = \varepsilon(\nu) = 0$. Define the automorphism ψ of F to fix $\sigma, \tau, x_0, x_2, x_4, \dots, x_d$ and

$$\psi(x_1) = x_1 \cdot w_c, \qquad \psi(x_3) = x_3 \cdot w_\mu\, w_\nu,$$

where $w_c, w_\mu, w_\nu \in P$ are elements with images $c\,\bar{x}_2$, $\mu\,\bar{x}_1$ and $\nu\,\bar{x}_2$ in M_1. As ψ is unipotent on M_1, it is an automorphism of F.

(4) The change of the relation. Modulo $P^{(3)}$ we have $\psi(r)\,r^{-1} \equiv E$ with $E \in M_2$ equal to

$$E = p\,c\,\bar{x}_2 + B(\bar{x}_1, \delta c\,\bar{x}_2) + B(c\,\bar{x}_2, \delta\bar{x}_1) + B(c\,\bar{x}_2, \delta c\,\bar{x}_2) + B(\bar{x}_2, \mu\,\bar{x}_1) + B(\bar{x}_2, \nu\,\bar{x}_2)$$

(here $s = 1$; as p is odd, $(uv)^p \equiv u^p v^p$). By the rule of (2) and the alternation of B, modulo $I_0 M_2$,

$$E \equiv p\,c\,\bar{x}_2 + B\big(\bar{x}_2, (c^\dagger\delta - c^\dagger\delta^\dagger + \mu)\bar{x}_1\big) + B\big(\bar{x}_2, (c^\dagger\delta c + \nu)\bar{x}_2\big) = p\,c\,\bar{x}_2 \equiv 0,$$

the last step because $c \in I_0$. So $E \in I_0 M_2 = \theta M_2$, and there is $\zeta \in M_2$ with $\theta\zeta = -E$. Let $z \in P^{(2)}$ lift ζ, and let ψ' be ψ with $x_0 \mapsto x_0 z$ instead of $x_0 \mapsto x_0$. By (1), $\psi'(r) \equiv r \pmod{P^{(3)}}$.

(5) The lifting. ψ' fixes $\mathcal{G}$ and preserves the normal closure of r and $P^{(3)}$ (and so does its inverse). Since $G_k = F/\langle\langle r\rangle\rangle$ and $V^{(3)}$ is the image of $P^{(3)}$, ψ' induces an automorphism of $G_k/V^{(3)}$ that is the identity on G_k/V. By Theorem 6.3 there is $\varphi \in \mathrm{Aut}(G_k)$ that coincides with it modulo $V^{(2)}$.

(6) Computing φ_+. On $V/V^{(2)}$, φ acts by $\bar{x}_1 \mapsto \bar{x}_1 + c\,\bar{x}_2$, $\bar{x}_3 \mapsto \bar{x}_3 + \mu\,\bar{x}_1 + \nu\,\bar{x}_2$, fixing the other $\bar{x}_i$ ($z \in P^{(2)}$ has no effect). Passing to $(V/V^{(2)})_{\mathcal{G}} \cong U_k^1/(U_k^1)^p \cong \Lambda/p\Lambda$ ($\bar{x}_i \mapsto \bar{y}_i$, as in the proof of Lemma 3.2) gives $\bar{y}_1 \mapsto \bar{y}_1 + \varepsilon(c)\bar{y}_2 = \bar{y}_1 + \bar{y}_2$ and $\bar{y}_3 \mapsto \bar{y}_3 + \varepsilon(\mu)\bar{y}_1 + \varepsilon(\nu)\bar{y}_2 = \bar{y}_3$, the other $\bar{y}_i$ being fixed. The action of φ_+ modulo p is this map (§2.2). □

6.4 Consequences

Corollary 6.6. P *If K is small, $d = e \geq 3$ is odd and $f = 1$, then case (ii) of Theorem 6.1 holds: some $\gamma \in \Gamma_K$ does not preserve ℓ_0, and for $e \mid m$ we have $S_K(m) = \mathfrak{m}^{m+1-e}$ and $a_K(m) = e - 1$.*

Proof. If $\ell_0 = \mathbb{F}_p\bar{y}_1$, the φ of Theorem 6.5 sends $\bar{y}_1$ to $\bar{y}_1 + \bar{y}_2 \notin \ell_0$. If $\ell_0 \neq \mathbb{F}_p\bar{y}_1$, the transvection in the proof of Proposition 6.2 moves ℓ_0. □

Theorem 6.7 (main theorem)**.** P *Let p be an odd prime and $K/\mathbb{Q}_p$ a finite extension with ramification index $e \leq p - 2$. Then for every integer m*

$$S_K(m) = \mathfrak{m}_K^{\mathrm{lo}(m)}, \qquad a_K(m) = (m-1) \bmod e.$$

In particular the images of $\mathcal{O}_K$ span the log-shell $\mathfrak{m}_K^{1-e}$, and if $e \geq 2$, then $v_K(\gamma(x)) = 1 - e$ for some $x \in \mathcal{O}_K$ and $\gamma \in \Gamma_K$.

Proof. The case $e = 1$ is trivial. For $e \geq 2$, the only case not covered by Theorem 5.1 is $d = e$ odd, $f = 1$ and $e \mid m$, which is Corollary 6.6 together with Theorem 6.1(ii) (here $\mathrm{lo}(m) = m + 1 - e$). □

Corollary 6.8 ($(\mathbb{Q}_p)_+$ is moved)**.** P *Let p be an odd prime and $k/\mathbb{Q}_p$ a finite extension of odd degree $d \geq 3$ with $s(k) = 1$ (not necessarily small; $s(k) = 1$ if $p \nmid e$). Then there is $\alpha \in \mathrm{Aut}(G_k)$ with $\alpha_+\big((\mathbb{Q}_p)_+\big) \cap (\mathbb{Q}_p)_+ = 0$.*

Proof. As d is odd, $\mu_p \not\subset k$; the proof of Lemma 3.2 goes through, and $y_1, \ldots, y_d$ is a $\mathbb{Z}_p$-basis of $\Lambda = \log U_k^1$. By [Kondo, Thm. 1.3], $W := \Lambda \cap \ker \mathrm{Tr} = \langle y_2, \ldots, y_d\rangle_{\mathbb{Z}_p}$. Write $(\mathbb{Q}_p)_+ \cap \Lambda = \mathbb{Z}_p u$ with u primitive and $u = c'y_1 + w'$ with $c' \in \mathbb{Z}_p$, $w' \in W$; as $\mathrm{Tr}(u) \neq 0$, $c' \neq 0$. (a) If $c' \in \mathbb{Z}_p^\times$, the φ of Theorem 6.5 gives $\varphi_+(u) \equiv u + c'y_2 \not\equiv u \pmod{p\Lambda}$. Since φ_+ preserves the trace, $\varphi_+(u) \in \mathbb{Q}_p u$ would force $\varphi_+(u) = u$, a contradiction. (b) If $c' \in p\mathbb{Z}_p$, then $w' \notin pW$, and some basis element v of a pair has $\langle w', v\rangle \in \mathbb{Z}_p^\times$. The transvection T_v of Proposition 4.4(b) fixes y_1, and $T_v(u) = u + \langle w', v\rangle\, v$; comparing coefficients of y_1 shows $T_v(u) \notin \mathbb{Q}_p u$. In both cases the two lines are distinct, so they meet in 0. □

Remark 6.9 (Kondo's question). P The φ of Theorem 6.5 satisfies $\varphi_+(y_1) \neq y_1$; this is the element whose existence in odd degree Kondo left open [Kondo, Rem. 1.10] (answered here for $s(k) = 1$; for $s(k) \geq 2$ see Remark 6.11). In the proof of Kondo's Theorem 1.9 (for even d_k and absolutely Galois k, a representation built from a Lubin–Tate character is not Aut-intrinsically Hodge–Tate) the parity of the degree enters only through the automorphism φ of [Kondo, Lemma 1.5], and the property used there is $\varphi_+((\mathbb{Q}_p)_+) \cap (\mathbb{Q}_p)_+ = 0$ [Kondo, proof of Lemma 1.5(2)]. The automorphism of Corollary 6.8 has this property, so the same proof applies to absolutely Galois k of odd degree $d \geq 3$ with $s(k) = 1$. The conclusion itself was proved by Kondo in a later paper, by an argument with Haar measures that does not produce explicit automorphisms [Kondo26, Thm. D]. What Corollary 6.8 adds is the existence of elements that move $(\mathbb{Q}_p)_+$ itself, and, modulo p, the line ℓ_0.

Remark 6.10. The φ of Theorem 6.5 is explicit only modulo $V^{(2)}$; its existence as an automorphism of G_k comes from Theorem 6.3. NSW use the same theorem to construct outer automorphisms of $G_{\mathbb{Q}_p}$ [NSW, pp. 421–423], and Nishio to construct finite subgroups acting on the pairs $(x_2, x_3), \ldots$ [Nishio26, Prop. 2.6]; those automorphisms fix x_0 and x_1, so the line of y_1 is not moved. Here x_1 is shifted towards x_2 at the third level; the commutator terms that this creates are absorbed by x_3, the p-th power term and the constraint from the trace by the condition $c \in I_0$ on the twisted augmentation, and the rest by the adjustment of x_0.

Remark 6.11. H For k of odd degree with $s(k) \geq 2$ (which occurs only when $p \mid e$; for example, the subfield of degree p of $\mathbb{Q}_p(\zeta_{p^2})$), Theorem 6.5 is not proved here. The same construction with the descending p^s-central series, together with the lifting theorem in the form of [Nishio26, Prop. 2.6], would presumably work, but this has not been checked. Version 1.0.0 of this note stated Theorem 6.3 without the condition on s, and Theorem 6.5, Corollary 6.8 and Remark 6.9 including this case. Since $s = 1$ for small K, Theorem 6.7 and Section 7 are not affected.

7 Application to Step (v) of IUT IV

7.1 Setting (summary of [Tiers])

S We use the notation of [Tiers, §2]. Let p be an odd prime, $j \in \{1, \ldots, \ell^*\}$ and $I = \{0, 1, \ldots, j\}$. For each $i \in I$ let v_i be a place above p (not necessarily distinct), $k_i = K_{v_i}$, $R_i = \mathcal{O}_{k_i}$, e_i its ramification index and π_i a uniformizer; all factors are assumed small, so that the different has order $\mathfrak{d}_i = (e_i - 1)/e_i$. Put $R_I = \bigotimes_{\mathbb{Z}_p} R_i$, let $(R_I)^\sim$ be its normalization, $T = R_I \otimes \mathbb{Q}_p$ and $\mathfrak{d}_I = \sum_i \mathfrak{d}_i$. The element $p^\lambda \in k_j$ ($\lambda \in \frac{1}{e_j}\mathbb{Z}$) sits in the factor j, and the region is $\Theta = p^\lambda \cdot (R_I)^\sim$. The holomorphic hull $\mathrm{hull}(X)$ of $X \subseteq T$ is the intersection of the sets $c \cdot (R_I)^\sim$ ($c \in T^\times$) containing X [IUT3, Rem. 3.9.5(i)]. The class $\mathfrak{A}$ of the actual (Ind1), (Ind2) consists of the maps $\bigotimes_i c_i\gamma_i$ with $c_i \in \mathbb{Z}_p^\times$ and $\gamma_i \in \Gamma_{k_i}$ [Tiers, Def. 2.3]. Put $U_{\mathfrak{A}} = \bigcup_{g \in \mathfrak{A}} g(\Theta)$.

[Tiers] shows: for the classes of maps that preserve the log-shells the hull is exactly $p^{\lceil\lambda\rceil - \mathfrak{d}_I} \cdot (R_I)^\sim$ [Tiers, Thm. 3.2]; $\mathrm{hull}(U_{\mathfrak{A}})$ is contained in it [Tiers, Thm. 4.2]; and equality holds if $\lambda \in \mathbb{Z}$ and every v_i satisfies condition (U), that some $\gamma \in \Gamma_{K_v}$ and $x \in \mathcal{O}_{K_v}$ have $|\gamma(x)| = |\pi_v|^{1-e_v}$ [Tiers, Def. 4.3, Thm. 4.4]. That note did not prove (U) for $e_v \geq 2$, and said nothing for $\lambda \notin \mathbb{Z}$ or for odd degree [Tiers, §4.1, §8].

Theorem 7.1. P *If all factors are small, then*

$$\mathrm{hull}(U_{\mathfrak{A}}) = p^{\lceil\lambda\rceil - \mathfrak{d}_I} \cdot (R_I)^\sim;$$

that is, the actual (Ind1), (Ind2) reach the exact value of [Tiers].

Proof. For $i \neq j$, Theorem 6.7 with $m = 0$ gives $x_i \in R_i$ and $\gamma_i \in \Gamma_{k_i}$ with $v_{k_i}(\gamma_i(x_i)) = \mathrm{lo}_{e_i}(0) = 1 - e_i$ (for $e_i = 1$ the identity will do). For the factor j put $m := e_j\lambda \in \mathbb{Z}$;

Theorem 6.7 gives $x \in \mathfrak{m}_{k_j}^m$ and γ_j with $v_{k_j}(\gamma_j(x)) = \mathrm{lo}_{e_j}(m)$. Write $x = p^\lambda x_j$ with $x_j \in R_j$. Then

$$y := x_0 \otimes \cdots \otimes x_{j-1} \otimes p^\lambda x_j \in p^\lambda R_I \subseteq \Theta,$$

and its image under $\bigotimes_i \gamma_i \in \mathfrak{A}$ is a pure tensor of nonzero elements, hence a unit of T whose order in every component (the ord of [Tiers, §2.1]) is

$$\sum_{i \neq j} \frac{1 - e_i}{e_i} + \frac{\mathrm{lo}_{e_j}(m)}{e_j} = -\sum_{i \neq j} \mathfrak{d}_i + \lceil \lambda \rceil - \mathfrak{d}_j = \lceil \lambda \rceil - \mathfrak{d}_I;$$

here we used $\mathrm{lo}_{e_j}(m) = e_j \lfloor (m-1)/e_j \rfloor + 1$ and $\lfloor (m-1)/e_j \rfloor = \lceil \lambda \rceil - 1$. A set $c \cdot (R_I)^\sim$ that contains a unit u contains $u \cdot (R_I)^\sim$ ($uc^{-1} \in (R_I)^\sim$, and $(R_I)^\sim$ is a ring), so $\mathrm{hull}(U_{\mathfrak{A}}) \supseteq p^{\lceil \lambda \rceil - \mathfrak{d}_I} \cdot (R_I)^\sim$. The reverse inclusion is [Tiers, Thm. 4.2]. □

Corollary 7.2. P *Condition (U) of [Tiers] holds at every small place.*

Proof. (U) says $a_{K_v}(0) = e_v - 1$, which is Theorem 6.7 (trivial for $e_v = 1$). □

Example 7.3. N The explicit initial Θ-datum D109 of [Cat] has $E_0 \colon y^2 = x(x-1)(x-a)$ over $F_0 = \mathbb{Q}(i)$, $a = \pi^{29}$ with $\pi = 10 + 3i$, $\ell = 7$, $F = F_0(\sqrt{-1}, E_0[15])$ and $K = F(E_0[7])$ [Cat, §3]. The program of [Walk] lists six places of $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$, above

$$p = 109,\ 21577,\ 6065177,\ 534775081,\ 1174770007439769673,\ 16450055877403724897.$$

At the place w of F_0 above 109 we have $\mathrm{ord}_w(a) = 29$, and at the other five $\mathrm{ord}_w(1-a) = 1$; so the reduction is split multiplicative over $F_{0,w} = \mathbb{Q}_p$ (the tangents at the node are $\pm\sqrt{-1}$, resp. ± 1, and all six primes are $\equiv 1 \bmod 4$), with $\mathrm{ord}_w(q) = 58$, resp. 2, prime to 105. By Tate's uniformization the completion of K is $K_v = \mathbb{Q}_p(\mu_{420}, q^{1/105})$, totally ramified of degree 105 over the unramified extension $\mathbb{Q}_p(\mu_{420})$. Hence $e = 105$, and f is the order of p modulo 420: $6, 12, 4, 1, 12, 4$ respectively. All six places are small. At five of them $[K_v : \mathbb{Q}_p]$ is even and Theorem 5.1(a) applies. At the place above $534775081 \equiv 1 \pmod{420}$, $K_v = \mathbb{Q}_p(q^{1/105})$ has degree 105 and is totally ramified; there Theorem 5.1 does not reach $e \mid m$, in particular condition (U) ($m = 0$), and Corollary 6.6 is used. The table of [Tiers, §5] has all factors at the bad place above 109, where $\lambda = 29j^2/7 \notin \mathbb{Z}$; by Theorem 7.1, its column "exact" is the value reached by the actual (Ind1), (Ind2).

Remark 7.4. [Tiers] gave as a sufficient condition for (U) that the vector η of a transvection $x \mapsto x + \mathrm{Tr}(x)\eta$ lie on the boundary of the log-shell [Tiers, Prop. 4.5]. Corollary 7.2 proves (U) without this condition: the proofs of Section 5 use transvections together with the maps that mix the pairs, not a single transvection, and in odd degree with total ramification the lifting of Section 6. Theorem 7.1 also covers $\lambda \notin \mathbb{Z}$ (always the case at bad places), about which [Tiers] said nothing. The container of Step (v) of IUT IV remains larger than the exact value by exactly a factor p when $\lambda \notin \mathbb{Z}$ [Tiers, Cor. 3.3]; this difference does not change the shape of the inequality of IUT IV.

8 Two types that are not small: the cyclotomic boundary and fields unramified over $\mathbb{Q}_2$

Sections 2–6 dealt with small K (p odd, $e \leq p-2$). In this section we determine $S_K(m)$ and $a_K(m)$ for two types that are not small (the remaining types are Question 10.3). The new tool is that the group of principal units that become p-th powers in the unramified extension of degree p is determined by the group. The proofs in §§8.1–8.3 do not use the lifting theorem of Section 6 (Proposition 8.13(a) in §8.4 uses Theorem 6.5). For $p = 2$ too, α_+ and Γ_K are defined as in §2.2 (Hoshi's reconstruction is stated for all mixed-characteristic local fields [Hoshi, §3]). For $n \geq 1$ write $U^{(n)} := 1 + \mathfrak{m}^n$ (so $U^1 = U^{(1)}$).

8.1 An invariant lattice from the unramified extension of degree p

Lemma 8.1. P *Let $K/\mathbb{Q}_p$ be a finite extension (p any prime), let L be the unramified extension of K of degree p, and put*

$$C_K := U_K^1 \cap (L^\times)^p,$$

the group of principal units of K that are p-th powers in L. Then $\gamma(\log C_K) = \log C_K$ for every $\gamma \in \Gamma_K$.

Proof. The inertia group I_K and the wild inertia group P_K of G_K are group-theoretic [Hoshi, Def. 3.5(iii), Prop. 3.6], so they are preserved by every continuous automorphism β of G_K. The subgroup $H := G_L$ is the inverse image of $p\widehat{\mathbb{Z}}$ under $G_K \to G_K/I_K \cong \widehat{\mathbb{Z}}$, so $\beta(H) = H$. In local class field theory the inclusion $\iota\colon K^\times \to L^\times$ corresponds to the transfer $G_K^{\mathrm{ab}} \to H^{\mathrm{ab}}$ [Hoshi, Lemma 1.7(3)]. The transfer commutes with isomorphisms of groups, so the multiplicative actions β_K, β_L that β and $\beta|_H$ induce on $K^\times$ and $L^\times$ (through the reciprocity maps) commute with ι. They preserve the principal units [Hoshi, Prop. 3.11(i)]. Let $u \in C_K$ with $\iota(u) = y^p$, $y \in L^\times$. Then y is a unit with $\bar{y}^p = 1$ in the residue field, so $y \in U_L^1$, and $\iota(\beta_K(u)) = \beta_L(y)^p \in (L^\times)^p$, that is, $\beta_K(u) \in C_K$. The same holds for β^{-1}, so $\beta_K(C_K) = C_K$. The map $\gamma = \beta_+$ of §2.2 is the action of β_K on U_K^1 transported by log, so $\gamma(\log C_K) = \log C_K$. □

Lemma 8.2. P *Assume $\mu_p \subset K$ and put $e_1 := e/(p-1)$ ($e_1 = e$ if $p = 2$). Then*

$$C_K = U^{(pe_1)}\,(U^1)^p.$$

In particular, if $e_1 = 1$ ($e = p-1$, or $p = 2$ and $e = 1$), then $(U^1)^p \subseteq U^{(p)}$, so $C_K = U^{(p)}$.

This is the classical description of the units that give unramified Kummer extensions (primary elements) [FV, Ch. IV, (1.4) and §1, Exercise 7]; we include a proof.

Proof. Put $\rho := \zeta_p - 1$ ($\rho = -2$ if $p = 2$). Then $v_K(\rho) = e_1$ and $p/\rho^{p-1} \equiv -1 \pmod{\mathfrak{m}}$. For $A \in \mathcal{O}$ let $u = 1 + \rho^p A \in U^{(pe_1)}$ and

$$F_A(X) := \frac{(1+\rho X)^p - u}{\rho^p} = X^p + \frac{p}{\rho^{p-1}}X - A + \sum_{j=2}^{p-1}\binom{p}{j}\rho^{j-p}X^j.$$

The middle coefficients have valuation $(j-1)e_1 > 0$, so F_A is a monic polynomial over $\mathcal{O}$, and its reduction to the residue field κ is the separable polynomial $X^p - X - \bar{A}$. This has a root in the extension of κ of degree p (the residue field of L), so by Hensel's lemma F_A has a root $\xi \in \mathcal{O}_L$, and $(1+\rho\xi)^p = u$. Hence $U^{(pe_1)} \subseteq C_K$; clearly $(U^1)^p \subseteq C_K$. Conversely, for $u \in C_K$, $u = y^p$ with $y \in L$, the map $\sigma \mapsto \sigma(y)/y \in \mu_p$ gives an injection of $C_K/(U^1)^p$ into $\mathrm{Hom}(\mathrm{Gal}(L/K), \mu_p)$, which has order p (it does not depend on the choice of the root since $\mu_p \subset K$; for elements of the kernel $y \in K$, and then $y \in U_K^1$). If A is chosen with $\mathrm{Tr}_{\kappa/\mathbb{F}_p}(\bar{A}) \neq 0$, the corresponding u is not a p-th power in K: a p-th root $w \in K$ would satisfy $v_K(w-1) \geq e_1$, so $w = 1 + \rho X$ with $X \in \mathcal{O}$, and $\bar{X}$ would be a root in κ. So the image of $U^{(pe_1)}$ is everything, and $C_K = U^{(pe_1)}(U^1)^p$. If $e_1 = 1$, then for $x \in \mathfrak{m}$ we have $v_K(px) \geq e+1 = pe_1$ and $v_K(x^p) \geq p$, so $(1+x)^p \in U^{(p)}$. □

Proposition 8.3. P *If $\mu_p \subset K$, every $\gamma \in \Gamma_K$ preserves $p\Lambda + \mathfrak{m}^{pe_1}$ and $\Lambda + \mathfrak{m}^{e_1}$.*

Proof. As $pe_1 > e/(p-1)$, $\log U^{(pe_1)} = \mathfrak{m}^{pe_1}$, so $\log C_K = p\Lambda + \mathfrak{m}^{pe_1}$ by Lemma 8.2. The claim follows from Lemma 8.1, $\mathbb{Q}_p$-linearity and $p^{-1}\mathfrak{m}^{pe_1} = \mathfrak{m}^{e_1}$. □

Remark 8.4. S Constructing a lattice from the group side by means of an auxiliary extension is of the same kind as Hoshi's group-theoretic reconstruction of $\mathfrak{m}^n$ for $\mathbb{Q}_3(\zeta_3)$, which uses that the log of the image of the norm from $k(\zeta_9)$ is $\mathfrak{m}^3$ [HoshiTop, Lemma 1.8(iv), Def. 2.9(ii)]. Here the unramified extension and the transfer (the inclusion) are used. Proposition 8.3 does not follow from the invariance of Λ (Proposition 2.3) alone.

8.2 The cyclotomic boundary

In this subsection p is an odd prime, $e = p-1$ and $\mu_p \subset K$. Then $K \supseteq \mathbb{Q}_p(\zeta_p)$ and $K/\mathbb{Q}_p(\zeta_p)$ is unramified, so $\pi := \zeta_p - 1$ is a uniformizer, and $d = (p-1)f$ is even. Let κ be the residue field.

Lemma 8.5. P

(a) $\mathfrak{m}^2 \subset \Lambda \subset \mathfrak{m}$ *and* $[\mathfrak{m} : \Lambda] = p$; $\Lambda/\mathfrak{m}^2$ *consists of the lifts of* $\pi \cdot \ker \mathrm{Tr}_{\kappa/\mathbb{F}_p}$.

(b) $\mathrm{Tr}(\Lambda) = p\mathbb{Z}_p$.

(c) The smallest valuation on $\Lambda \setminus \{0\}$ *is* 2 *if* $f = 1$ *and* 1 *if* $f \geq 2$ *(if* $f = 1$, *then* $\Lambda = \mathfrak{m}^2$*).*

(d) $\gamma(\mathfrak{m}) = \mathfrak{m}$ *for every* $\gamma \in \Gamma_K$.

Proof. (a) As $2 > e/(p-1) = 1$, $\log U^{(2)} = \mathfrak{m}^2$. For $z \in \mathcal{O}$ the terms $(\pi z)^n/n$ of $\log(1+\pi z)$ lie in $\mathfrak{m}^2$ for $2 \leq n \leq p-1$ and for $n > p$, so $\log(1+\pi z) \equiv \pi(z + (\pi^{p-1}/p)z^p) \pmod{\mathfrak{m}^2}$, and $\pi^{p-1}/p \equiv -1 \pmod{\mathfrak{m}}$. The image of $z \mapsto z - z^p$ on the residue field is $\ker \mathrm{Tr}_{\kappa/\mathbb{F}_p}$ (Artin–Schreier), which gives (a). (b) $K/\mathbb{Q}_p$ is tamely ramified and $\mathrm{Tr}(\mathfrak{m}^j) = p^{\lceil j/e \rceil}\mathbb{Z}_p$, which is $p\mathbb{Z}_p$ for $j = 1, 2$. (c) follows from (a). (d) By Lemma 8.2, $C_K = U^{(p)}$ and $\log C_K = \mathfrak{m}^p = p\mathfrak{m}$ (as $e = p-1$); apply Lemma 8.1. □

Lemma 8.6. P *Choose the Jannsen–Wingberg generators so that condition (0) of Proposition 1.1 of Hoshi–Nishio holds (it suffices to take the integers* g, h *there larger than* p *[HN, Prop. 1.1 and its proof]), and put* $y_i := \log \mathrm{rec}_K^{-1}(x_i)$ *(*$0 \leq i \leq d$*). Then* $y_1, \ldots, y_d$ *is a* $\mathbb{Q}_p$*-basis [HN, Lemma 1.3], there is* $z \in \mathbb{Q}_p y_1$ *with* $\mathbb{Z}_p y_0 + \mathbb{Z}_p y_1 = \mathbb{Z}_p z$, *and*

$$\Lambda = \mathbb{Z}_p z \oplus \bigoplus_{i=2}^{d} \mathbb{Z}_p y_i, \qquad \mathrm{Tr}(z) = \mathrm{Tr}(y_i) = 0 \ (i \geq 3), \qquad \mathrm{Tr}(y_2) \in p\mathbb{Z}_p^{\times}.$$

Proof. P_K is normally generated by $x_0, \ldots, x_d$, and its image in G_K^{ab} is $\mathrm{rec}_K(U^1)$, so $\Lambda = \sum_{i=0}^{d} \mathbb{Z}_p y_i$ (conjugates become equal in the abelianization, and the sum is closed since it is finitely generated). Abelianizing the relation and taking log gives $h\, y_0 = p\, y_1$ with $h \in \mathbb{Z}_p$ (here the s of §6.2 is 1, as $p \nmid e$). As $y_1 \neq 0$, $h \neq 0$, so $y_0 \in \mathbb{Q}_p y_1$, and $\mathbb{Z}_p y_0 + \mathbb{Z}_p y_1$ has rank 1, so it is $\mathbb{Z}_p z$ for some z. The sum is direct by the linear independence of $y_1, \ldots, y_d$. As d is even, $\ker \mathrm{Tr} = \langle y_1, y_3, \ldots, y_d \rangle_{\mathbb{Q}_p}$ [Kondo, Thm. 1.3] (whose proof uses this basis property). By Lemma 8.5(b), $\mathrm{Tr}(y_2)$ generates $p\mathbb{Z}_p$. □

Condition (0) cannot be dropped: for $g = h = 1$ the abelianized relation reads $p\bar{x}_1 = 0$, and $y_1 = 0$.

Lemma 8.7. P *Put* $V := \Lambda/p\Lambda$, *write* $[x]$ *for the class of* $x \in \Lambda$, *and let* $\tau := \mathrm{Tr}/p \bmod p$ *(a linear form on* V*) and* $R := \mathfrak{m}^p/p\Lambda \subset V$. *Then* R *is preserved by* Γ_K, *and* $R = \mathbb{F}_p[z]$.

Proof. From $\mathfrak{m}^p = p\mathfrak{m}$ and $[\mathfrak{m} : \Lambda] = p$ we get $\dim R = 1$, and from $\mathrm{Tr}(\mathfrak{m}^p) = p^2\mathbb{Z}_p$ we get $\tau(R) = 0$. Since $\mathfrak{m}$ and Λ are preserved (Lemma 8.5(d)), so is R. Put $U := \mathrm{span}_{\mathbb{F}_p}([y_3], \ldots, [y_d])$; then $\ker \tau = \mathbb{F}_p[z] \oplus U$. If $d = 2$, then $U = 0$ and we are done. If $d \geq 4$, by Kondo Γ_K contains elements that fix y_1, y_2 and act on $\langle y_3, \ldots, y_d \rangle$ as $\mathrm{Sp}_{d-2}(\mathbb{Z}_p)$ [Kondo26, Lemma 3.3] (they also fix z, as $y_0 \in \mathbb{Q}_p y_1$). Their reductions modulo p act transitively on $U \setminus \{0\}$, so U has no invariant line. The projection $\ker \tau \to U$ along $\mathbb{F}_p[z]$ is equivariant for this subgroup, so the image of R is 0, and $R = \mathbb{F}_p[z]$. □

Lemma 8.8. P *If* $2 \leq r \leq e$, *then* $S_K(r) = \Lambda$.

Proof. If $d = 2$, then $p = 3$, $f = 1$, $r = 2$ and $\mathfrak{m}^2 = \Lambda$. Let $d \geq 4$. We have $p\Lambda \subset \mathfrak{m}^p \subset \mathfrak{m}^r \subset \mathfrak{m}^2 \subset \Lambda$. Let N be the $\mathbb{F}_p$-subspace of V spanned by the Γ_K-orbit of the image of $\mathfrak{m}^r$. Then $N \supseteq R = \mathbb{F}_p[z]$. As $\mathrm{Tr}(\mathfrak{m}^r) = p\mathbb{Z}_p$, N contains an element x with $\tau(x) \neq 0$, that is, whose coefficient β of $[y_2]$ is nonzero. The automorphism given by the mixing word M of §4.3 on the first two pairs $(x_1, x_2), (x_3, x_4)$ (Lemmas 4.1 and 4.3) fixes x_0 and x_1, acts on log by $y_2 \mapsto y_2 + y_1 + y_4$, $y_3 \mapsto y_3 - y_1 - y_4$ ($M^{\mathrm{ab}} = T_{a+d}$), and fixes the other y_i. If η is the coefficient of $[y_3]$ in x, then $Mx - x = (\beta - \eta)([y_1] + [y_4])$. If $\beta = \eta$, apply first an element that multiplies the pair (y_3, y_4) by -1 (an element of $\mathrm{Sp}_{d-2}(\mathbb{Z}_p)$); then the coefficient of the difference is $2\beta \neq 0$ (p odd). Hence $[y_1] + [y_4] \in N$. As $[y_1] \in \mathbb{F}_p[z] \subset N$, also $[y_4] \in N$ (it does not matter whether the α with $y_1 = \alpha z$ is divisible by p). As $\mathrm{Sp}_{d-2}(\mathbb{F}_p)$ acts transitively on $U \setminus \{0\}$, $U \subset N$. Since N contains $\ker \tau = \mathbb{F}_p[z] \oplus U$ and an element with $\tau \neq 0$, $N = V$. Together with $p\Lambda \subset \mathfrak{m}^r \subseteq S_K(r)$ this gives $S_K(r) = \Lambda$. □

Theorem 8.9 (the cyclotomic boundary)**.** P *Let p be an odd prime and $K/\mathbb{Q}_p$ a finite extension with $e = p - 1$ and $\mu_p \subset K$. Write $m = ne + r$ with $1 \leq r \leq e$. Then*

$$S_K(m) = \mathfrak{m}^m \quad (r = 1), \qquad S_K(m) = p^n\Lambda \quad (2 \leq r \leq e),$$

and hence

$$a_K(m) = \begin{cases} 0 & (r = 1), \\ r - 2 = (m - 2) \bmod e & (r \geq 2,\ f = 1), \\ r - 1 = (m - 1) \bmod e & (r \geq 2,\ f \geq 2). \end{cases}$$

In particular the images of $\mathcal{O}_K$ span the log-shell $p^{-1}\Lambda$, and the image of some integral element reaches its boundary.

Proof. For $r = 1$, $\mathfrak{m}^m = p^n\mathfrak{m}$ is preserved (Lemma 8.5(d)). For $r \geq 2$, $\mathfrak{m}^m = p^n\mathfrak{m}^r \subseteq p^n\Lambda$ and $p^n\Lambda$ is preserved, so $S_K(m) \subseteq p^n\Lambda$, with equality by Lemma 8.8 and $\mathbb{Q}_p$-linearity. The value of $a_K(m)$ follows from the smallest valuation $ne + w$ on $p^n\Lambda$, with w as in Lemma 8.5(c); that the valuation of an image actually takes this value follows by the argument of Section 5. The ring $\mathcal{O}_K = \mathfrak{m}^0$ is the case $n = -1$, $r = e$, and $S_K(0) = p^{-1}\Lambda$ is the log-shell $(2p)^{-1} \log \mathcal{O}^\times = p^{-1}\Lambda$. □

Example 8.10. P For $\mathbb{Q}_3(\zeta_3)$ ($e = 2$, $f = 1$), $a_K(m) = 0$ for every m, in agreement with Hoshi's result that $\mathcal{O}$ and all $\mathfrak{m}^n$ are group-theoretic (Remark 8.12). For $\mathbb{Q}_5(\zeta_5)$, $a_K(m) = 0, 0, 1, 2$ for $m = 1, 2, 3, 4$. For the unramified extensions of $\mathbb{Q}_3(\zeta_3)$ with $f \geq 2$, $a_K(m)$ is 0 for odd m and 1 for even m.

8.3 Fields unramified over $\mathbb{Q}_2$

Theorem 8.11. P *Let $K/\mathbb{Q}_2$ be a finite unramified extension. Then $\gamma(\mathfrak{m}^m) = \mathfrak{m}^m$ for every $\gamma \in \Gamma_K$ and every integer m. Hence $S_K(m) = \mathfrak{m}^m$, $a_K(m) = 0$, and Γ_K preserves the ring of integers $\mathcal{O}_K$.*

Proof. Here $e = 1$, $\mu_2 \subset K$ and $e_1 = 1$. By Lemma 8.2, $C_K = U^{(2)} = 1 + 4\mathcal{O}_K$, and $\log C_K = 4\mathcal{O}_K = \mathfrak{m}^2$ (as $2 > e/(p-1) = 1$). By Lemma 8.1, $\mathfrak{m}^2$ is preserved; the claim follows from $\mathfrak{m}^m = 2^{m-2}\mathfrak{m}^2$ and $\mathbb{Q}_2$-linearity. □

Remark 8.12. S For $K = \mathbb{Q}_2$ and $K = \mathbb{Q}_3(\zeta_3)$ (and for unramified K with p odd) the log-shell equals the ring of integers, and Hoshi reconstructs $\mathcal{O}_K$ and all $\mathfrak{m}^n$ group-theoretically [HoshiTop, Thm. A(ii), Lemma 1.8(i)–(iv), Def. 2.9, Summary 2.11(ii)]. For K unramified over $\mathbb{Q}_2$ of degree at least 2, the log-shell does not equal $\mathcal{O}_K$ [HoshiTop, Thm. A(ii)], but by Theorem 8.11 every automorphism preserves $\mathcal{O}_K$. We do not know whether in this case G_K has automorphisms that do not come from field automorphisms. For $p = 2$ and $\sqrt{-1} \notin K$, no

presentation of G_K by generators and relations is known [Kondo, Rem. 2.22]. Theorem 8.11 should be read as a constraint on any automorphism. For 2-adic fields containing $\sqrt{-1}$, Nishio constructs automorphisms that do not come from field automorphisms [Nishio25, Thm. 1.6].

8.4 Unions of images

Instead of the module spanned by the images of a lattice, consider the union of the images itself,

$$A_K(m) := \bigcup_{\gamma\in\Gamma_K} \gamma(\mathfrak{m}^m) \;\subseteq\; S_K(m).$$

The holomorphic hull of IUT sees only the spanned module, so this does not affect Section 7.

Proposition 8.13. P

(a) Let K be small and $d \geq 3$ odd. If $f \geq 2$, then $A_K(m) = \mathfrak{m}^{\mathrm{lo}(m)}$ for every m. If $f = 1$ (so $d = e$), then $A_K(m) = \mathfrak{m}^{\mathrm{lo}(m)}$ for $e \nmid m$, and for $m = ne$

$$A_K(ne) = p^{n-1}\big(\{x \in \Lambda : \mathrm{Tr}(x) \notin p^2\mathbb{Z}_p\} \cup p\Lambda\big).$$

(b) On the cyclotomic boundary (the setting of Theorem 8.9) write $m = ne+r$ with $1 \leq r \leq e$. If $r = 1$, then $A_K(m) = \mathfrak{m}^m$; if $r \geq 2$ and $(f, r) \neq (1, e)$, then $A_K(m) = p^n\Lambda$; if $f = 1$ and $r = e$, then

$$A_K(m) = p^n\big(\{x \in \Lambda : \mathrm{Tr}(x) \notin p^2\mathbb{Z}_p\} \cup \mathfrak{m}^p\big).$$

(c) In the case $m = ne$ of (a), and in the case $f = 1$, $r = e$, $d \geq 4$ of (b), $A_K(m)$ is not an additive group, and its Haar measure relative to $S_K(m)$ is $1-1/p+p^{-d}$ and $1-1/p+p^{1-d}$ respectively (for example $101/125$ for $\mathbb{Q}_5(\zeta_5)$).

Proof. Let $q\colon \Lambda \to V := \Lambda/p\Lambda$ be the quotient map. For a $\mathbb{Z}_p$-submodule B with $p\Lambda \subseteq B \subseteq \Lambda$ we have $\bigcup_\gamma \gamma(B) = q^{-1}\big(\bigcup_\gamma \gamma(q(B))\big)$: if $q(x) = \gamma q(b)$ with $b \in B$, then $x - \gamma(b) \in p\Lambda$ and $\gamma^{-1}(x - \gamma(b)) \in p\Lambda \subseteq B$, so $x = \gamma(b + \gamma^{-1}(x - \gamma(b))) \in \gamma(B)$. So it suffices to determine the orbits in V.

(a) We have $V = \mathbb{F}_p\bar{y}_1 \oplus \bar{W}$ with $\bar{W} = W/pW$, and $\tau := \mathrm{Tr}\,/\,\mathrm{Tr}(y_1) \bmod p$ is preserved by Γ_K. The elements of Γ_K that fix y_1 and act on W as $\mathrm{Sp}_{d-1}(\mathbb{Z}_p)$ [Kondo26, Lemma 3.3] fix $\bar{y}_1$ and act transitively on $\bar{W} \setminus \{0\}$. The φ of Theorem 6.5 ($s = 1$ for small K) maps $\bar{y}_1$ to $\bar{y}_1 + \bar{y}_2$. So the orbits in V are $\{0\}$, $\bar{W} \setminus \{0\}$ and the sets $\tau^{-1}(t)$ $(t \neq 0)$. For $1 \leq r \leq e$ the subspace $I_r := q(\mathfrak{m}^r)$ has dimension $f(e+1-r)$ and $\tau(I_r) = \mathbb{F}_p$. If the dimension is at least 2, I_r meets every orbit, the union is V, and $A_K(r) = \Lambda$. If it is 1 ($f = 1$, $r = e$), then $I_e = \mathbb{F}_p\bar{p}$ and the union is $\{0\} \cup \{\tau \neq 0\}$. The general case follows from $\mathfrak{m}^{ne+r} = p^n\mathfrak{m}^r$.

(b) $R = \mathbb{F}_p[z]$ and τ are preserved by Γ_K (Lemma 8.7). Use coordinates (t, u) on $V/R = \mathbb{F}_p[y_2] \oplus U$. If $d = 2$, then $U = 0$, and as τ is preserved the orbits are points. If $d \geq 4$, the M of the proof of Lemma 8.8 acts modulo R by $(t, u) \mapsto (t, u + (t - u_3)[y_4])$, where u_3 is the coefficient of $[y_3]$ in u. The map $u \mapsto u - u_3[y_4]$ is a transvection in the direction $[y_4]$, an element of Sp_{d-2}, so composing with its inverse gives the translation $(t, u) \mapsto (t, u + t[y_4])$, and conjugating by Sp_{d-2} gives $(t, u) \mapsto (t, u + tv)$ for every $v \in U$. So the orbits in V/R are $\{0\}$, $U \setminus \{0\}$ and $\{t\} \times U$ $(t \neq 0)$. We have $q(\mathfrak{m}^r) \supseteq R$, and the intersection of $q(\mathfrak{m}^r)/R$ with $\ker\tau$ has dimension $f(e+1-r)-1$. If this is at least 1, the union is V; if it is 0 ($f = 1$, $r = e$), the union is $R \cup \{\tau \neq 0\}$, and it cannot be larger since all of Γ_K preserves R and τ. The inverse image of R is $\mathfrak{m}^p$.

(c) In V the set $\{\tau \neq 0\}$ has $(p-1)p^{d-1}$ elements. Add the element 0 in (a) and the p elements of R in (b), and divide by p^d. The union is not an additive group: in (a), $\bar{y}_1$ and $-\bar{y}_1 + w$ $(0 \neq w \in \bar{W})$ lie in it but their sum does not; in (b) the same holds for $[y_2]$ and $-[y_2] + u$ $(0 \neq u \in U)$. □

Remark 8.14. S Mochizuki writes that for any $\mathbb{Z}_p$-automorphism ϕ of the log-shell $\mathcal{I}_k$ one has $\phi(U(\mathcal{O}_k, n)) \subseteq \bigcup_{i=-s}^{s} U(\mathcal{O}_k, n+i)$, where $U(\mathcal{O}_k, n)$ is the set of elements of $p^n\mathcal{O}_k$ that are not in $p^{n+1}\mathcal{O}_k$, and s depends only on k [EssLgc, Ex. 3.5.1(ii)]. Proposition 8.13 determines exactly the union of the images of a lattice under the automorphisms of the actual Galois group.

9 Relation to earlier work

- S The ring of integers is not group-theoretic. Hoshi states that there are automorphisms of G_L that do not preserve the ring of integers of $\overline{L}$, and that no functorial group-theoretic algorithm can reconstruct $(\mathcal{O}_k)_+$ uniformly for all mixed-characteristic local fields [Hoshi, Rem. 4.3.1(ii)(iii)], [HoshiTop, Rem. 2.11.2]. Hoshi also shows that the log-shell equals the ring of integers exactly when p is odd and k is unramified, or $k \cong \mathbb{Q}_2$ or $\mathbb{Q}_3(\zeta_3)$ [HoshiTop, Thm. A(ii)], and in these cases reconstructs the ring of integers and all $\mathfrak{m}^n$ group-theoretically [HoshiTop, Def. 2.9, Summary 2.11(ii)]. The present note measures, for each small K, how far the ring of integers is moved.

- S Types that are not small (Section 8). The case $\mathbb{Q}_3(\zeta_3)$ of Theorem 8.9 and the case $\mathbb{Q}_2$ of Theorem 8.11 are contained in Hoshi's reconstruction above, and constructing lattices with an auxiliary extension is of the same kind as [HoshiTop, Lemma 1.8(iv)] (Remark 8.4). For the cyclotomic boundary with $p \geq 5$ or $f \geq 2$, and for fields of degree at least 2 unramified over $\mathbb{Q}_2$, we have not found an earlier treatment. The description of the units that give unramified Kummer extensions is classical [FV, Ch. IV, (1.4) and §1, Exercise 7]. For 2-adic fields containing $\sqrt{-1}$, Nishio constructs automorphisms that do not come from field automorphisms [Nishio25, Thm. 1.6]. For unions of images, Mochizuki writes a bound in the form of a union for any $\mathbb{Z}_p$-automorphism of the log-shell [EssLgc, Ex. 3.5.1(ii)] (Remark 8.14).

- S Upper bounds. Hoshi's valuation with an indeterminacy [HoshiTop, Prop. 1.10(ii)] bounds the drop of the valuation (Remark 2.6). The bound of Proposition 2.5 comes from the invariance of Λ and is of the same kind; the claim here is that it is always attained.

- S Traces and transvections. Hoshi and Nishio prove the preservation of traces and give unipotent induced maps for abelian extensions of even degree [HN, Lemma 2.3(ii), Thm. 1.5]. Kondo describes the kernel of the trace by the generators of the Jannsen–Wingberg presentation [Kondo, Thm. 1.3], gives in even degree an automorphism moving $(\mathbb{Q}_p)_+$ [Kondo, Lemma 1.5], and shows that $\mathrm{Sp}_{2g}(\mathbb{Z}_p)$ lies in the image of the induced maps [Kondo, proof of Thm. 2.19], [Kondo26, Lemma 3.3]. The maps and arguments of Sections 4 and 5 are of the same kind; this note combines them on the integral basis to determine the movement of the lattices.

- S The question in odd degree. Kondo wrote that it was not clear whether an α with $\alpha_+(y_1) \neq y_1$ exists in odd degree [Kondo, introduction, Rem. 1.10]. A later paper of Kondo proves, for nontrivial finite Galois extensions including those of odd degree, that the Lubin–Tate representations are not Aut-intrinsically Hodge–Tate, by an argument with Haar measures [Kondo26, Thms. C, D]; that paper does not construct elements moving y_1 or $(\mathbb{Q}_p)_+$, does not refer to Remark 1.10, and notes that explicit automorphisms are not obtained [Kondo26, remark after Thm. D, Rem. 3.7]. Theorem 6.5 shows that the element of Remark 1.10 exists in odd degree when $s(k) = 1$ (which always holds if $p \nmid e$).

- S The lifting theorem and the presentation. We use the presentation of Jannsen and Wingberg [JW], the computation of the linear part of the relation in Section 2 of their paper, and Wingberg's lifting theorem [Wi, Satz 2], [NSW, Thm. 7.5.15]. NSW construct

nontrivial outer automorphisms of $G_{\mathbb{Q}_p}$ with it [NSW, pp. 421–423]. Nishio uses the same lifting to construct finite subgroups of $\mathrm{Aut}(G_k)$ acting on the pairs $(x_2, x_3), \ldots$ [Nishio26, Prop. 2.6, Thm. 2.8]; Nishio states the theorem with the descending p^s-central series, which agrees with the form in NSW in the case $s = 1$ used here (the remark after Theorem 6.3). Those automorphisms fix x_0 and x_1, so they do not move the line of y_1, and the paper does not consider the ring of integers or $(\mathbb{Q}_p)_+$. Kumpitsch's Jannsen–Wingberg automorphisms and combinatorial Dehn twists [Ku, Prop. 3.2.4, Thm. 7.5.5] and Gropper's arithmetic Dehn twists [Gr] do not concern the movement of the ring of integers either. For $p = 2$, Nishio studies the outer automorphism groups of the absolute Galois groups of 2-adic fields [Nishio25].

- S IUT. The author's note [Tiers] determined the exact hull in Step (v) and gave condition (U) for the actual (Ind1), (Ind2) to reach it; Section 7 continues it. Mochizuki writes that the valuation data "is precisely the data that is 'deformed' by automorphisms of k that do not arise from field automorphisms" [AbsTopIII, §I3], describes (Ind1) as arising from automorphisms of G_k that do not in general preserve the ring structure [Alien, §3.7(i)], writes that "automorphisms of the $\mathbb{Z}_p$-module I_k only give rise to bounded discrepancies in the orders of nonzero elements of $\mathcal{O}_k$" [EssLgc, Ex. 3.5.1], and, in an e-mail quoted by Joshi, that such automorphisms "can act in a fashion that fails to preserve differents, discriminants" and other quantities [JoshiAnab, Appendix 16]. Scholze and Stix note: "Such isomorphisms exist for Galois groups of p-adic fields, but this did not seem to enter the discussion in a critical way" [SS, fn. 4]. Mochizuki's formalization report mentions "a small discrepancy (arising from (Ind1,2,3) + hull)" [Fm26, p. 11]. Dupuy and Hilado treat (Ind1), (Ind2) as acting through the automorphisms of the $\mathbb{Z}_p$-lattices of the log-shells [DH1, p. 17, §4.8]; lana-agents/iut uses the group `tensorAut` of factorwise automorphisms of the log-shells, which contains the image of Ism and is described there as a deliberate enlargement (`Iut/Concrete/LocalConstruct/IndAut.lean`, 11 October 2026), and states that for $\lambda = 0$ the hull is exactly the container of IUT IV (`Iut/WallEquiv/VcI.md`, 13 October 2026) [lana]. Minamide and Tsujimura prove $\mathrm{Ism}(G_K) = \mathbb{Z}_p^\times$ [MT, Thm. F, Thm. 7.4], and write "Theorem F implies that (Ind2) is relatively small" [MT]. Takkun-kohinata/IUT_LEAN states, in a comment of `Thm311.lean`, that (Ind1), (Ind2) preserve volume and so do not contribute to the volume estimate of the hull (our paraphrase) [Takkun]; this concerns each image. Zeng computes the index by which the hull enlarges the integral structure of a tensor product of rings of integers [Zeng]. Joshi reads (Ind1) as anabelomorphy [JoshiIII, §8.11.2], a reading different from the one used here.
- P What is added here: (1) for small K, the determination of $S_K(m)$ and $a_K(m)$ for every m; (2) in odd degree $d \geq 3$ with $s(k) = 1$, the existence of an automorphism moving y_1 (the element of Kondo's Remark 1.10), and of automorphisms moving $(\mathbb{Q}_p)_+$ and, modulo p, the line ℓ_0; (3) for Step (v) of IUT IV, the removal of the hypotheses $\lambda \in \mathbb{Z}$ and (U) of [Tiers]: at small places the actual (Ind1), (Ind2) reach the same hull as the factorwise automorphisms of the log-shells; (4) an invariant lattice determined by the unramified extension of degree p, and the movement of the lattices on the cyclotomic boundary and for fields unramified over $\mathbb{Q}_2$ (Section 8); (5) the unions of images (Proposition 8.13).

10 Remaining questions

What is determined here is the module spanned by the images of the lattices $\mathfrak{m}_K^m$ (and, in some cases, the union of the images). For the group of induced maps itself, and outside the scope of this note, the following questions remain.

Question 10.1 (the group of induced maps)**.** Let p be an odd prime, $K/\mathbb{Q}_p$ a finite extension with $\mu_p \not\subset K$, $\Lambda = \log U_K^1$, and

$$T_K := \{\, g \in \mathrm{GL}_{\mathbb{Z}_p}(\Lambda) \mid \mathrm{Tr} \circ g = \mathrm{Tr} \,\}.$$

By Proposition 2.3, $\Gamma_K \subseteq T_K$ (restricting to Λ). Is $\Gamma_K = T_K$? If not, is Γ_K an open subgroup of T_K? At least, do their images in $\mathrm{GL}(\Lambda/p\Lambda)$ agree?

Remark 10.2 (what is known)*.* We record what is known.

- P For small K, by Theorem 6.7, Γ_K and T_K give the same modules spanned by the images of $\mathfrak{m}_K^m$ for every m (both $\mathfrak{m}_K^{\mathrm{lo}(m)}$: T_K preserves Λ, so the upper bound is the same, and $\Gamma_K \subseteq T_K$ gives the lower bound). The movement of the lattices does not distinguish the two.
- P In odd degree $d = 2g+1 \geq 3$ with $s(K) = 1$, with $W := \Lambda \cap \ker \mathrm{Tr}$, the image of Γ_K in $\mathrm{GL}(\Lambda/p\Lambda)$ satisfies

  $$(W/pW) \rtimes \mathrm{Sp}_{2g}(\mathbb{F}_p) \;\subseteq\; \mathrm{Im} \;\subseteq\; (W/pW) \rtimes \mathrm{GL}(W/pW),$$

  where W/pW is the group of translations $\bar{y}_1 \mapsto \bar{y}_1 + w$ (identity on W/pW). The left inclusion follows from Kondo's $\mathrm{Sp}_{2g}(\mathbb{Z}_p) \subseteq \Gamma_K$ [Kondo26, Lemma 3.3] and the conjugates under Sp_{2g} of the φ of Theorem 6.5 (the translation $\bar{y}_1 \mapsto \bar{y}_1 + \bar{y}_2$ modulo p); the right one from the preservation of the trace.
- S Kondo proves the lower bound $\dim \Gamma_K \geq 2g^2 + g$ for the dimension as a p-adic Lie group [Kondo26, Lemma 3.3]; the dimension of T_K is $d^2 - d$.
- H The image modulo p presumably reduces, by the method of Section 6 (automorphisms at the third level and Theorem 6.3), to a computation of the automorphism group of a twisted form over $\mathbb{F}_p[[\mathcal{G}]]$. For example, the map moving the pair (x_2, x_3) by $\bar{x}_2 \mapsto a\bar{x}_2$, $\bar{x}_3 \mapsto (a^\dagger)^{-1}\bar{x}_3$ seems to give an element outside Sp on W/pW when $\varepsilon(a) \neq \chi_0(a)$, in which case the image would be the whole right-hand side. This has not been checked.
- S For Demuškin groups, every automorphism that preserves the cup product and the Bockstein comes from a group automorphism [Ku, Thm. 6.3.2], and Kumpitsch conjectures a surjectivity of the same kind for G_K with $\mu_{p^t} \subseteq K$ [Ku, p. 17]. Question 10.1 is an analogue on the units for $\mu_p \not\subset K$.

Question 10.3 (larger ramification)**.** For $p = 2$ with $K/\mathbb{Q}_2$ ramified, or for odd p and $e \geq p$, determine $S_K(m)$ and $a_K(m)$.

The case $e = p - 1$ is settled by Remark 5.5 ($\mu_p \not\subset K$) and Theorem 8.9 ($\mu_p \subset K$), and the case of fields unramified over $\mathbb{Q}_2$ by Theorem 8.11. In the remaining cases an upper bound again follows from the invariance of $\Lambda = \log U_K^1$, but in general $\Lambda \neq \mathfrak{m}_K$ and the form changes. If $\mu_p \subset K$, the invariant lattice $\Lambda + \mathfrak{m}^{e_1}$ of Proposition 8.3 ($e_1 = e/(p-1)$) can also be used for the upper bound. For odd p and odd degree, Theorem 6.5 can be used when $s(K) = 1$. For $p = 2$ one has to restart from the tools of Sections 3–6 (the Jannsen–Wingberg presentation for odd primes); see [Nishio25] for outer automorphisms in the 2-adic case. We also do not know whether, for K unramified over $\mathbb{Q}_2$ of degree at least 2, G_K has automorphisms that do not come from field automorphisms (Remark 8.12).

Question 10.4 (an explicit automorphism)**.** For odd $d \geq 3$, give an automorphism of G_k with $\alpha_+(y_1) \neq y_1$ explicitly, as words in the Jannsen–Wingberg generators.

The φ of Theorem 6.5 is determined only modulo the second level, and its existence as an automorphism of G_k comes from the lifting theorem (Theorem 6.3), which gives existence and not the form of the words. Kondo also notes, for a different purpose (automorphisms that destroy the Hodge–Tate property), that explicit automorphisms are not obtained [Kondo26, Rem. 3.7].

Question 10.5 ($s \geq 2$)**.** For $k/\mathbb{Q}_p$ of odd degree $d \geq 3$ with $s(k) \geq 2$, is there $\alpha \in \mathrm{Aut}(G_k)$ with $\alpha_+(y_1) \neq y_1$?

This amounts to checking the expectation of Remark 6.11.

11 Limitations

- The statements about lattices concern only small K, K with $e = p - 1$ (Remark 5.5, Theorem 8.9), and K unramified over $\mathbb{Q}_2$. The cases $p = 2$ with K ramified and p odd with $e \geq p$ (wild ramification) are not treated.
- Γ_K itself is not determined, and the φ of Theorem 6.5 is explicit only modulo the second level. Theorem 6.5 is proved only for $s(k) = 1$ (Remark 6.11).
- Theorem 6.5 uses as known inputs the presentation of Jannsen and Wingberg with the computations of Section 2 of their paper, and Wingberg's lifting theorem (NSW (7.5.15)). The proof of the lifting theorem [Wi] has not been checked here; we rely on the form stated in NSW, who note that the statement in [Wi] was not correct as stated. This note uses only the case $s = 1$, in which the form in NSW agrees with that of [Nishio26, Prop. 2.6] (the remark after Theorem 6.3).
- The check in Lean (Section 12) is relative: the cited results and some classical facts are hypotheses, and Lean does not check that these hypotheses state the cited results correctly. Lemma 2.2(c)(d), Remark 5.5, Lemma 8.5, the assembly of the automorphism of G_k in Theorem 6.5 (the computation in the presentation and the lifting), and the passage to the holomorphic hull in Theorem 7.1 are not in Lean. The author's own reading of the proofs is still in progress.

12 Machine verification

Many of the P statements are checked in Lean 4 (v4.30.0) with Mathlib (v4.30.0). The files are in the folder `log-shell-edge/lean` of the deposit (Section 13). What is checked is that each statement follows, by the proof given here, from the hypotheses in the right-hand column of Table 2. The cited results (S) and the classical facts about local fields that are not in Mathlib are hypotheses of the theorems (fields of structures), not axioms. Lean does not check that these hypotheses state the cited results correctly. Every theorem depends on no axioms other than `propext`, `Classical.choice` and `Quot.sound`, and no `sorry` and no `native_decide` is used; the file `Verify.lean` of the deposit prints the axioms of the main results. The README of the folder gives the names of the Lean theorems.

In most files the field K is replaced by a model: a $\mathbb{Q}_p$-vector space V with the lattices $L_j = \mathfrak{m}^j$ ($L_{j+e} = pL_j$, strictly decreasing), a basis y of $\Lambda = L_1$, the linear form $\tau = \mathrm{Tr}\,/p$, and a subgroup Γ of linear automorphisms (the maps α_+). The multiplication of K, the group G_K and the reciprocity map are not in Lean; what the proofs use about them enters as hypotheses. Only Lemma 2.2(a)(b) is proved for an actual complete ultrametric normed field.

Not in Lean are Lemma 2.2(c)(d), Remark 5.5, Lemma 8.5, Examples 7.3 and 8.10, Remark 10.2, and the parts marked "not in Lean" in Table 2.

Statement	Checked in Lean	Hypotheses
Lemma 2.2(a)(b)	$\mu_p \not\subset K$; U^1 is torsion-free; log is a bijection $U^1 \to \mathfrak{m}$ that preserves norms and is a homomorphism	$\rho^e = \|p\|$ (ρ the norm of a uniformizer), $e \le p-2$, discreteness ($\|x\| < 1$ implies $\|x\| \le \rho$)
Proposition 2.5	$S_K(m) \subseteq \mathfrak{m}^{\mathrm{lo}(m)}$ (hence $a_K(m) \le (m-1) \bmod e$), $S_K(m+e) = p\,S_K(m)$, $a_K(m+e) = a_K(m)$	(Γ1)
Lemma 3.2	the computation of the $\mathbb{Z}_p$-module modulo the relation (if it is torsion-free, the coefficient is a unit and $y_1, \dots, y_d$ are independent)	the p-part of the abelianization of G_K is this module (Theorem 3.1 and the reciprocity map)
Lemma 3.4	all	Kondo's description of the kernel of the trace; Lemma 2.2(d)
Lemma 4.1	that the automorphism descends through the presentation to X	that α_+ acts on Λ by the matrix of the abelianization (through the reciprocity map and log)
Lemmas 4.2, 4.3	all (identities in free groups, abelianizations)	none
Proposition 4.4; the map M of Lemma 8.8	derived from the conclusion of Lemma 4.1	the conclusion of Lemma 4.1: an automorphism of the free group fixing z_1 and the boundary word is realized in Γ_K by the matrix of its abelianization
Lemmas 5.2–5.4, Theorems 5.1, 6.1, Proposition 6.2 (the direction giving (ii))	all	$\Lambda = \mathfrak{m}$ with its $\mathbb{Z}_p$-basis, (Γ1), (Γ2), the dimensions of $\mathfrak{m}^r/p\mathfrak{m}$, the element of Lemma 2.2(d), Kondo's description of the kernel of the trace, the conclusion of Lemma 4.1; for odd degree and total ramification also $\mathfrak{m}^e/p\mathfrak{m} = \mathbb{F}_p\bar{p}$
Theorem 6.5	the identities in the group ring in steps (2)–(4) and (6) of the proof	Theorem 6.5 itself; the computation in the presentation, $\theta R = I_0$, Proposition 6.4 and the lifting (Theorem 6.3) are not in Lean
Corollaries 6.6, 6.8, 7.2, Theorem 6.7, Remark 6.9	all	the hypotheses above, and Theorem 6.5 (for odd degree and total ramification, and for Corollary 6.8 and Remark 6.9)
Theorem 7.1	the identity of orders, and the hull of a set containing a unit	the setting of [Tiers]; the passage to the container and the holomorphic hull is not in Lean
Lemmas 8.1, 8.2	the group-theoretic cores (maps commuting with the inclusion, a counting)	the identification with the groups of principal units, and the transfer
Proposition 8.3, Theorem 8.11	all	the invariance of $\log C_K$ (Lemmas 8.1, 8.2)
Lemma 8.6	the computation of the module	[HN, Prop. 1.1, Lemma 1.3]
Lemmas 8.7, 8.8, Theorem 8.9, Proposition 8.13	all	the lattice facts of Lemma 8.5, the basis of Lemma 8.6, the transvections of [Kondo26, Lemma 3.3], the conclusion of Lemma 4.1; for Proposition 8.13(a) the hypotheses on small fields and Theorem 6.5

Table 2: What is checked in Lean, and what is taken as a hypothesis.

It is also checked in Lean that the maps that the proofs put into Γ (those of Proposition 4.4, M, the map of Theorem 6.5, Kondo's transvections) preserve the trace; so these maps do not contradict (Γ2). The conclusion of Lemma 4.1 is not checked in this way for every automorphism of the free group.

The automorphisms Z, H, J, M of the free group in §4.3, the two module lemmas used in Lemmas 3.2 and 5.2, and the logarithm series over $\mathbb{Q}_p$ and over complete ultrametric normed fields are adapted from an earlier, unpublished Lean development of the author (the comments were rewritten, the namespaces renamed and unused statements removed).

13 Reproducibility

The folder `log-shell-edge/` of the repository `https://github.com/turahigashi/iut-explicit-initial-theta` (tag `v0.27.0`), also deposited with this note, contains the Lean files of Section 12 (folder `lean`, with `Verify.lean` and a README), and scripts in Python (standard library only, except that the last one uses the program of [Walk], which needs sympy) with their outputs. The checks do not replace the proofs.

- `check_free_group.py`: the identities of Section 4 in the free group (Z, H, J and the swap of pairs with their inverses, the preservation of the boundary words, $M(a) = a$) and the matrices of the abelianizations ($Z^{\mathrm{ab}} = T_{d-b}$, $M^{\mathrm{ab}} = T_{a+d}$, the formula for U_d).
- `check_lemma52.py`: for $d = 2, 4, \ldots, 12$ and random integral x, the determinant of the d vectors of Lemma 5.2 is $-u^d$.
- `check_group_ring.py`: steps (2)–(4) and (6) of the proof of Theorem 6.5 in finite models. For eleven metacyclic groups of order 4 to 27 (abelian and non-abelian) and all their nontrivial characters χ_0 over $\mathbb{F}_p$ (25 pairs), it checks that e' is a central idempotent, that $\theta R = R\theta = I_0$, the adjoint rule, that $\varepsilon(c) = 1$, $c \in I_0$ and $\varepsilon(\mu) = \varepsilon(\nu) = 0$, and, for random δ, that E lies in $I_0 M_2$ and that $\theta\zeta = -E$ is solvable, where $M_2 = M_1 \oplus \Lambda^2 M_1$ models the second level of a free pro-p group; E itself is nonzero in every trial, so without the adjustment of x_0 the relation is not preserved at the third level.
- `check_boundary_lattices.py`: the lattice facts of Section 8 (Lemma 8.5(a)–(c), the case $e_1 = 1$ of Lemma 8.2, $\log U^{(2)} = \mathfrak{m}^2$, the values of a_K in Theorem 8.9 and Example 8.10, and the log-shell of fields unramified over $\mathbb{Q}_2$) for $\mathbb{Q}_p(\zeta_p)$ ($p = 3, 5, 7$), the unramified extensions of degree 2 and 3 of $\mathbb{Q}_3(\zeta_3)$, the unramified quadratic extension of $\mathbb{Q}_5(\zeta_5)$ and the fields unramified over $\mathbb{Q}_2$ of degree $1, 2, 3$, by exact integer arithmetic modulo p^{10}.
- `check_boundary_orbits.py`: the orbit computations of Lemmas 8.7, 8.8 and Proposition 8.13, in models over $\mathbb{F}_p$ generated only by the elements whose action is stated in the text (the transvections of Sp, M and φ): for (b), $(p, d) = (3, 2), (3, 4), (5, 4), (3, 6), (7, 6)$ and every value of the α with $y_1 = \alpha z$; for (a), $(p, d) = (3, 3), (5, 3), (7, 3), (5, 5), (7, 5)$.
- `example_d109.py`: the data of Example 7.3, read from the program `iut_compute.py` of [Walk] (folder `iut-walkthrough/program` of the same repository).

Changes in version 1.1.0

Version 1.0.0 of this note is dated 8 November 2026 and has the DOI 10.5281/zenodo.24050445. Version 1.1.0 (9 November 2026) differs from it as follows.

(1) Correction of scope. Version 1.0.0 stated Theorem 6.3 (Wingberg's lifting theorem) without a condition on $s(k)$, and Theorem 6.5, Corollary 6.8 and Remark 6.9 for every k of odd degree. The form stated in NSW uses the descending p-central series and cannot be used when $s(k) \geq 2$ (the remark after Theorem 6.3). The proof works when $s(k) = 1$,

which holds whenever $p \nmid e$, and these statements are now restricted to that case; the case $s(k) \geq 2$ is Remark 6.11 and Question 10.5. Since $s = 1$ for small K, the main theorem (Theorem 6.7) and Section 7 are unchanged.

(2) A new Section 8 on two types that are not small: an invariant lattice from the unramified extension of degree p, the cyclotomic boundary, fields unramified over $\mathbb{Q}_2$, and unions of images; with two new scripts.

(3) A new Section 12 on the check in Lean 4; the Lean files are in the deposit.

(4) Accordingly the abstract, Section 1, the relation to earlier work, the remaining questions, the limitations, the reproducibility section, the disclosure and the bibliography ([FV] added; for [HN] the revised version of June 2022 is used) were revised. The former Sections 8–10 are now Sections 9–11, and the former Section 11 is now Section 13.

Disclosure

The question studied here arose in the author's discussions with generative AI tools, and the author decided to make it the subject of this note. The proofs were found with generative AI tools (large language models) under the author's direction, and these tools were also used for the computations, the formalisation in Lean, the literature search and the writing. Section 12 says what is checked in Lean and under which hypotheses; the correspondence between these hypotheses and the cited results is not checked by Lean. The author's own reading of the proofs is still in progress. The author takes full responsibility for the note.

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